
How Finely to Group? Learned Partitions for Calibrating LLM Confidence across Elicitation Procedures

Anonymous Author
Anonymous Institution

Abstract

Language models report confidence that depends on how they are asked: rewording a prompt or reversing a scale changes which answers look confident and breaks calibrations made under the original procedure. We ask whether grouping question-answer pairs by a frozen representation computed before elicitation provides calibration structure across elicitation procedures, and how finely to group under a limited label budget. Across three open models, two factual question-answering datasets and synthetic reliability models, groups improve on recalibrating the confidence report only when reliability differs between groups and labels are plentiful (thousands of labelled questions, and even then the gain is small); otherwise their estimation cost outweighs the benefit, and a direct predictor on the same features does better. Groups do not shrink conformal prediction sets, and they do not remove the dependence on the procedure, because the forecast still reads the report. A monotone map between two procedures’ reports, fitted without labels, carries a calibration across procedures on average but not within every group. Calibration should therefore be checked, and reported, per prompt and scale.

1 INTRODUCTION

Asking a language model how confident it is costs one extra prompt, needs no access to weights, and returns a number a person can read; systems use such reports to abstain, to escalate to a human, or to trigger re-

trieval (Kadavath et al., 2022; Tian et al., 2023; Xiong et al., 2024). The report is calibrated to correctness under the procedure used to elicit it — a prompt, a scale, a legend that maps letters to percentages — and procedures change: prompts are edited, scales are reversed, templates are reworded.

A motivating observation. A model reports confidence for a fixed candidate answer by choosing a letter from A to L, read through a legend ($A = 0\% \dots L = 100\%$). Reverse the legend and decode through the reversed legend, and a calibration made under one legend should still hold under the other. It does not (Figure 1a, Table 1): on 7,189 PopQA questions, conformal sets calibrated under the forward legend cover 0.49 for Mistral-7B under the reversed legend and 0.50 under a reworded prompt, and 0.76 for Llama-3.1-8B under reversal, against a target of 0.90; a trained reporter falls from 0.90 to 0.20; other models become needlessly conservative. The reports themselves reshuffle: on 817 TruthfulQA questions a typical question moves 190–240 rank positions under reversal, against 239 if the two rankings were independent, and only 20–24% stay in their confidence fifth (20% by chance).

Three questions. Two remedies suggest themselves. One recalibrates the report separately under each procedure, which costs labels per procedure. The other groups question-answer examples by a frozen representation computed *before* elicitation — such groups cannot move when the procedure changes — and forecasts correctness from group and procedure together. The registered question of this paper is when the second provides useful calibration structure beyond the first, and when the cost of estimating more groups outweighs the benefit. The groups do not move with the procedure, but a forecast built on them still reads the report, so whether they help *across* procedures is a separate question; we find that they do not, and that what carries a calibration across procedures is a map between the reports. Answering the registered question requires three steps:

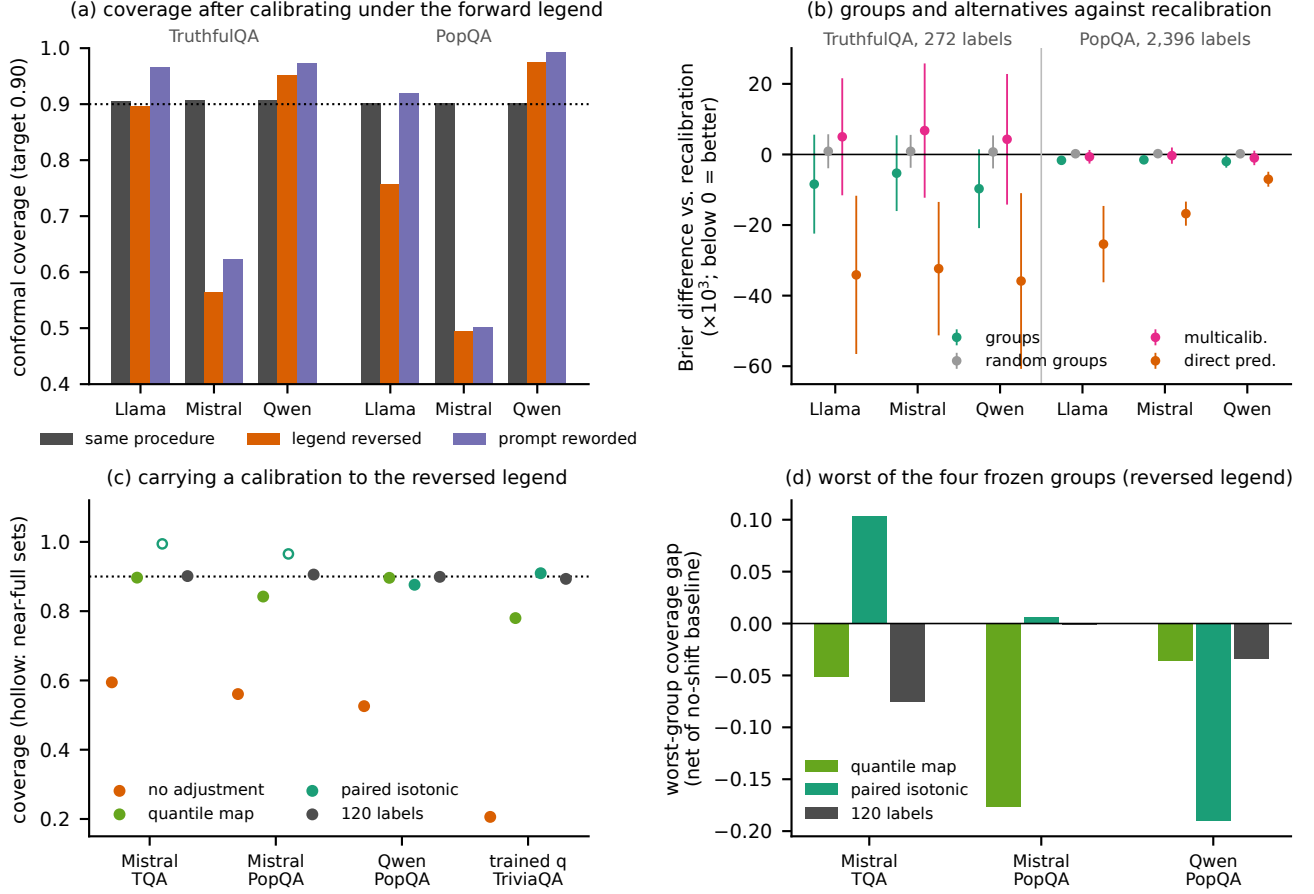


Figure 1: Overview. (a) Conformal coverage ($\alpha = 0.1$, target dotted) when a calibration made under the forward legend is deployed under the same legend, the reversed legend or a reordered prompt (three frozen models; mean over 500 random splits; Table 1). (b) Brier difference against procedure-specific recalibration ($\times 10^3$; below zero: better) at the largest label budget, for learned groups, random groups, multicalibration and a direct predictor on the same representation, with Nadeau–Bengio-corrected 95% intervals. (c) Coverage after carrying the calibration to the reversed legend: no adjustment, two label-free maps fitted on unlabelled questions elicited both ways, and recalibration on 120 labelled questions under the new legend; hollow markers mark near-full sets (at least 1.9 of 2 labels). (d) Coverage gap of the worst of the four frozen groups for the same arms, net of the same statistic with no procedure change; below zero, a group is left short.

1. **Which groupings license a decision?** A grouping must be computable when the decision is taken, must not be selected on the outcome, and must be used under the elicitation it was calibrated in (Section 3).
2. **How finely to group under a label budget?** Coarse groups hide reliability differences, fine groups leave too little data to estimate them (Section 5; GRANULARITY, SYNTHETIC-RELIABILITY).
3. **Can a calibration be carried across procedures without new labels?** Unlabelled questions can be elicited under both procedures, so the coupling of the two reports is observable and a

transport map can carry one to the other (Proposition 4; LABEL-FREE-TRANSFER).

Contributions. (i) A stage-indexed criterion for which groupings license a decision, synthesising known validity results, and a label-free transport proposition: the coverage change from mapping one procedure’s reports onto another’s is bounded by masses computable from unlabelled questions, and vanishes when the change is monotone (Section 3). (ii) The granularity trade-off for conformal coverage with frozen groups: cell-averaged error depends on counts only (Corollary 5), the pointwise error splits into quantisation and estimation (Proposition 6), and farthest-first cells admit a finite-sample bound (Lemma 7, Theorem 8); for forecasts by group and procedure the split

is a heuristic that the experiments test. (iii) An answer to the registered question: against procedure-specific recalibration, multicalibration and a direct predictor on the same features, groups help only where reliability varies between groups and labels are plentiful (on PopQA, one of fifteen pre-declared tests at 2,396 labels; a gain of about 1–2% of the Brier score), the best granularity grows with the budget, a tuned direct predictor does better throughout, groups do not shrink conformal sets, and groups do not protect calibration across a change of procedure. (iv) A transport phenomenon, studied in first experiments: a monotone map between paired reports repairs a calibration on average, exactly when the change is monotone, but not per group (LABEL-FREE-TRANSFER). Criteria were fixed in dated notes before each run, except where marked post hoc (Appendix A); POPQA-ELICIT and three other experiments were added after the TruthfulQA family returned no significant group effect, and pooled with it (27 tests) no group-versus-recalibration test survives Holm correction.

2 SETUP

2.1 Stages of information

Each unit is an independent draw of

$$W = (X, A, Y), \\ \mathcal{I}_0 = X, \quad \mathcal{I}_1 = (X, A), \quad \mathcal{I}_2 = (X, A, Y),$$

where X is the input (prompt), A is any randomness realised by the system before the outcome is revealed (e.g. a sampled answer and its decoding seed), and Y is the outcome (e.g. correctness). A decision is *taken at stage t* if it may use $\mathcal{I}_t$ but nothing later. The stage- t nonconformity score is $S = s(\mathcal{I}_t, Y)$ and the prediction set is $C(\mathcal{I}_t) = \{y : s(\mathcal{I}_t, y) \leq q_G\}$ for a cell index G .

2.2 Bins

A bin is a cell indexed by a stored exemplar. The general stage- t binning is Definition 18 in Appendix G; every result in Section 3 holds for it.

Definition 1 (Exemplar cells). Given a feature map ϕ and exemplars $e_1, \dots, e_M$, all fitted on D_{fit} only, $\gamma(x) = \arg \min_j \|\phi(x) - e_j\|$ (ties to the lower index). The balanced variant uses power cells $\arg \min_j (\|\phi(x) - e_j\|^2 - w_j)$ with weights w fitted on D_{fit} . A residual cell 0 collects queries outside every fitted radius (Proposition 14). After fitting, the cell of a query depends on that query alone (the *write-free* condition).

2.3 Elicitation procedures and forecasters

Each unit is $W = (X, \Pi, A, Y)$: the input, the elicitation procedure Π (legend, prompt wording), the system’s own randomness, and the outcome. The report is $V = V_\Pi(X, A)$; a group $G = \gamma(\phi(X))$ is computed from a frozen representation ϕ before any confidence prompt, so G does not depend on Π . Correctness forecasts are logistic in report, group and procedure,

$$\text{logit } \mathbb{P}(Y = 1 \mid V, G, \Pi = \pi) = a_\pi + b_\pi \text{logit } V + c_{G, \pi}, \quad (1)$$

with $c \equiv 0$ for procedure-specific recalibration R , $c_{G, \pi}$ free for the group forecaster G_M , and $c_{G, \pi} = c_G$ for offsets shared across procedures. For a fitted forecaster $\hat{f}$ and $\eta^* = \mathbb{P}(Y = 1 \mid V, Z, \Pi)$ with Z the full frozen representation (which does not depend on M), $\mathbb{E}(\hat{f} - Y)^2 = \mathbb{E}[\eta^*(1 - \eta^*)] + \mathbb{E}(\hat{f} - \eta^*)^2$ exactly, and $\mathbb{E}(\hat{f} - \eta^*)^2 \leq 2\mathbb{E}(f^* - \eta^*)^2 + 2\mathbb{E}(\hat{f} - f^*)^2$ with f^* the best model of the form (1). The approximation term falls as groups resolve reliability differences; the estimation term grows roughly as the number of parameters, $|\Pi|(M + 1)$ per n_{lab} labels (a heuristic, not a bound). This is the forecast-side analogue of the quantisation–estimation split of Section 5; GRANULARITY tests it. Theorem 8 concerns conformal pointwise coverage, not these forecasts.

2.4 Honest split and calibration rule

D_{fit} , D_{cal} , D_{test} are disjoint and sampled at the independent unit (question id for LLM data; one variant per question). The score, feature map, exemplars and cell rule are learned on D_{fit} alone. For cell j with n_j calibration scores, $t := 1 - \alpha$, $k_j = \lceil (n_j + 1)t \rceil$, and

$$q_j = \begin{cases} S_{(k_j)}^{(j)} & \text{if } k_j \leq n_j \text{ and } n_j \geq n_{\min}, \\ +\infty & \text{otherwise (full set).} \end{cases}$$

The floor $n_{\min}$ is the same in every arm; test points in below-floor cells are retained and the infinite-set mass is reported.

3 WHICH GROUPINGS LICENSE A DECISION

Theorem 2 (Per-cell validity). *Fix D_{fit} , hence s and γ . Let the calibration units and the test unit be exchangeable and independent of D_{fit} , and let $G = \gamma(\mathcal{I}_t)$. Then for every cell j ,*

$$\mathbb{P}\{Y_{\text{test}} \in C(\mathcal{I}_{t, \text{test}}) \mid G_{\text{test}} = j\} \geq 1 - \alpha,$$

and if scores are a.s. distinct and $q_j < \infty$, the conditional coverage given n_j is at most $1 - \alpha + 1/(n_j + 1)$.

Proposition 3 (Stage measurability). *(A restatement of Mondrian validity, Vovk, 2012, and of selection applied identically to calibration and test units, Bao et al., 2024, Thm. 1; the content is the stage indexing.) If the decision is taken at stage t , the binning γ is $\mathcal{I}_t$ -measurable and frozen on D_{fit} , the score is $s(\mathcal{I}_t, Y)$, and calibration and test units are retained by the same fixed $\mathcal{I}_t$ -measurable event E (possibly trivial), then the guarantee of Theorem 2 holds for the stage- t decision, conditional on E .*

Outcome selection breaks coverage (Example 11, Appendix F): calibrating on the answerable set and testing on the full stream gives coverage pt , where p is the answerable share, in the model of that example; answer-computed confidence licenses post-generation decisions only (Remark 13).

The stage criterion. We call Proposition 3 together with its three failure modes the *stage criterion*: a grouping licenses a decision only if it is computable from what is known at that decision, is not selected on the outcome (Example 11), and is applied under the elicitation convention it was calibrated in (Remark 13; ELICITATION-SHIFT).

Transporting a calibration across procedures. The stage criterion says a calibration fitted under procedure A is not licensed under B. Because a question can be elicited under both procedures without being labelled, the coupling of (V_A, V_B) is observable, and a label-free map T can carry B’s reports back to A’s scale. Let C_A be the conformal set computed from V_A with A’s forecaster, thresholds and frozen group, and C_T the set computed in the same way from $T(V_B)$.

Proposition 4 (Label-free transport). *(a) For any map T , $|\mathbb{P}(Y \in C_T) - \mathbb{P}(Y \in C_A)| \leq \mathbb{P}(C_T \neq C_A)$, and the same holds within each frozen group. The right-hand side, the crossing mass, depends only on (V_A, V_B, G) , so it can be estimated from unlabelled questions elicited under both procedures. One-sidedly, conditionally on the calibration data and on T , $\mathbb{P}(Y \in C_T) \geq \mathbb{P}(Y \in C_A) - \ell$ with the label-loss mass $\ell = \mathbb{P}(C_A \not\subseteq C_T)$, which again needs no labels; averaging over the calibration draw and using $\mathbb{P}(Y \in C_A) \geq 1 - \alpha$ gives $\mathbb{P}(Y \in C_T) \geq 1 - \alpha - \mathbb{E}\ell$. Both statements assume the outcome Y is the same under both procedures, which holds when the question and candidate are fixed. If $\hat{\ell}$ is the label-loss rate on m unlabelled questions independent of the calibration data and of the fold that fits T , then $\ell \leq \hat{\ell} + \sqrt{\log(1/\delta)/(2m)}$ with probability at least $1 - \delta$ given those, so the coverage under B is at most $\hat{\ell} + \sqrt{\log(1/\delta)/(2m)}$ below that under A; a union bound covers several maps or groups. (b) If $V_B = g(V_A)$ almost surely for a strictly monotone g , the population isotonic regression of V_A on V_B*

(in the direction that fits) equals g^{-1} , so $C_T = C_A$ and the guarantee of Theorem 2, per frozen group, transfers to B without labels. If g is increasing and the laws are continuous, the quantile map $F_A^{-1} \circ F_B$, the monotone optimal-transport map between the two marginals, also equals g^{-1} . (c) For continuous laws with finite second moments, the observed coupling of (V_A, V_B) is comonotone, i.e. (b) holds with g increasing, if and only if its squared-cost transport equals the optimal value W_2^2 between the two marginals; the gap between the realised and optimal transport costs measures the departure from a monotone change.

Groups computed before elicitation do not move with Π , which is what lets the per-group statements in (a)–(b) be used across procedures.

4 ESTIMANDS: POINTWISE VERSUS CELL LEVEL

Our primary estimand is the pointwise (conditional-on- X) coverage gap.

Fix D_{fit} and D_{cal} . Let $h(x) = F_x(q_{\gamma(x)})$, with F_x the conditional score CDF, $p_j = \mathbb{P}(\gamma(X) = j)$ and $C_j = \mathbb{E}[h(X) \mid \gamma(X) = j]$.

- pointwise gap: $R_{\text{pt}} = \mathbb{E}_X[(h(X) - t)^2]$;
- cell MSCE: $R_{\text{cell}} = \sum_j p_j (C_j - t)^2$;
- spread: $\max_j C_j - \min_j C_j$;
- concentrated undercoverage: $\max_{j: q_j < \infty} (t - C_j)_+$, eligible only if the full-set test mass is at most 1%.

Corollary 5 (Cell-averaged MSCE is blind to heterogeneity). *An aggregation of the per-cell Beta law (Vovk, 2012; Ding et al., 2023). Assume continuous within-cell score CDFs and condition on the counts (n_j) with $n_{\min} \leq n_j$ and $k_j \leq n_j$ for all j . Then $C_j \sim \text{Beta}(k_j, n_j + 1 - k_j)$ and*

$$\mathbb{E}[R_{\text{cell}} \mid (n_j)] = \sum_j p_j \left[\frac{k_j(n_j+1-k_j)}{(n_j+1)^2(n_j+2)} + \left(\frac{k_j}{n_j+1} - t \right)^2 \right],$$

which depends on the counts only, not on the conditional law of S given X . In particular, refinement that reduces within-cell heterogeneity cannot lower it; for $n_j \approx n_{\text{cal}} p_j$ large, it is $\approx t(1-t)M/n_{\text{cal}}$, increasing in M .

5 DECOMPOSITION AND THE NUMBER OF CELLS

Proposition 6 (Exact decomposition).

$$R_{\text{pt}} = \underbrace{\mathbb{E}_X[(h(X) - C_{\gamma(X)})^2]}_{\text{quantisation}} + \underbrace{\sum_j p_j (C_j - t)^2}_{R_{\text{cell}}}.$$

The quantisation term is evaluated at the *realised* thresholds; it is not an independent bias term.

Assumption 1 (Smoothness and geometry). (i) $\sup_s |F_x(s) - F_{x'}(s)| \leq L\|x - x'\|^\beta$, $\beta \in (0, 1]$. (ii) Every positive-mass cell has diameter at most $KM^{-1/d}$.

Lemma 7 (Cell diameters for farthest-first exemplars). *Let P_X be supported on a compact $S \subset \mathbb{R}^d$ with density $\geq c > 0$ there and $\text{Vol}(B(x, r) \cap S) \geq \kappa \omega_d r^d$ for $x \in S$, $r \leq r_0$, where ω_d is the volume of the unit ball (e.g. a convex body or $[0, 1]^d$). Let the exemplars be the first M points of a farthest-first traversal of D_{fit} ($n = n_{\text{fit}}$ i.i.d. points) and the cells nearest-exemplar cells. Put $a = c\kappa\omega_d$, fix $s > 1$, let $K = 2(\max(2, \text{diam } S/r_0) + 1)a^{-1/d}$, and let n be large enough that $\delta_n/2 \leq r_0$, where $\delta_n = 2(s \log n / (an))^{1/d}$. With probability at least $1 - 2^d n^{1-s} / (s \log n)$, simultaneously for all $M \leq n / (2^d s \log n)$, every cell intersected with S has diameter at most $KM^{-1/d}$. For $U[0, 1]^d$: $K = 6$ ($d = 1$), 6.77 ($d = 2$).*

Proof in Appendix B; scope (it covers farthest-first, not k -means, exemplars, and at $d = 16$ admits no $M \geq 2$) in Appendix F.

Theorem 8 (Upper bound for farthest-first cells). *Use nearest-exemplar cells without a residual cell. Assume Assumption 1(i), the conditions of Lemma 7, a density bounded above by $\bar{c}$, D_{cal} independent of D_{fit} , and a floor $n_{\min} \geq t/\alpha$. Let $b = c\kappa/(4^d \bar{c})$. Let $c'' = c\kappa/(4^d s \bar{c})$ and assume $\frac{1}{4}(\bar{c}\omega_d M)^{-1/d} \leq r_0$. For $2 \leq M \leq \min\{c''n_{\text{fit}}/\log n_{\text{fit}}, b n_{\text{cal}}/(2n_{\min})\}$,*

$$\begin{aligned} \mathbb{E}[R_{\text{pt}}] &\leq L^2 K^{2\beta} M^{-2\beta/d} + t(1-t) \frac{M}{n_{\text{cal}} + 1} \\ &\quad + \frac{3M^2}{b(n_{\text{cal}} + 1)^2} + \alpha^2 e^{-b n_{\text{cal}}/(8M)} + \frac{2^d n_{\text{fit}}^{1-s}}{s \log n_{\text{fit}}}. \end{aligned}$$

The terms are quantisation, Beta variance (no balance needed), order-statistic rounding, floor activation, and failure of the diameter event. Taking $M \asymp n_{\text{cal}}^{d/(d+2\beta)}$ gives $O(n_{\text{cal}}^{-2\beta/(d+2\beta)})$ provided $n_{\text{fit}}/\log n_{\text{fit}} \gtrsim n_{\text{cal}}^{d/(d+2\beta)}$ and s is large enough that $n_{\text{fit}}^{1-s}/\log n_{\text{fit}} = O(n_{\text{cal}}^{-2\beta/(d+2\beta)})$.

The rate itself is not new: it is the hard-partition analogue of kernel-localized conformal prediction under the same Hölder condition (Conrad et al., 2026; Min et al., 2026), going back to the partition-based method of Lei and Wasserman (2014), which gives the corresponding efficiency rate. What the hard partition adds is the exact finite-sample variance (Corollary 5), explicit floor and n_{fit} terms, and Remark 15.

A pre-asymptotic remark explaining why finite grids grow faster than this rate (Remark 15), the tight interior-optimum conjecture (Conjecture 16) and the

forecast (Brier) analogue of the decomposition (Proposition 17) are in Appendix F.

6 EXPERIMENTS

Setup and data. A frozen model (Llama-3.1-8B, Mistral-7B, Qwen2.5-7B) is shown a question and one candidate answer (teacher-forced) and reports its confidence that the candidate is correct by choosing a letter A–L; the report V is the expected value of its letter distribution decoded through the legend in force. The three procedures are the forward legend, the reversed legend and a reworded prompt. Data: TruthfulQA (Lin et al., 2022b) (817 questions) and POPQA-ELICIT, the same elicitations on PopQA (Mallen et al., 2023) with a same-relation distractor (7,189 questions after removing 188 alias collisions); TriviaQA (Joshi et al., 2017) answers of Llama-3.1-8B with trained question-only and answer-aware reporters; and synthetic reliability models. One candidate per question is chosen by a label-blind hash, so Y is its correctness and conformal sets are subsets of $\{0, 1\}$ (size 2 means abstaining); every procedure’s forecaster is fitted on the same labelled questions, and an unlabelled third fits the PCA and the groups. Overlapping random splits get Nadeau–Bengio-corrected intervals (Nadeau and Bengio, 2003); groups versus recalibration is a pre-declared Holm family (Holm, 1979); every logistic forecaster tunes its penalty by inner cross-validation, with the report’s coefficient penalised a hundred times less. Details are in Appendix H.

6.1 Elicitation changes break calibration

Ranks. Under reversal a typical question moves 190–240 rank positions, against 239 under independent rankings and 139–169 under a reworded prompt, and only 20–24% of questions stay in their confidence fifth (28–34% under rewording). The reports are weak on TruthfulQA (AUROC 0.40–0.60). Rank intervals, transport plans between the two legends and the quintile transitions are in Appendix H.

Coverage. (ELICITATION-SHIFT; criteria fixed before the run; 500 random thirds of the 817 TruthfulQA questions, and of the 7,189 POPQA-ELICIT questions, one label-blind candidate per question, $M = 4$). Calibrating and deploying under the same elicitation is valid for all three models on both datasets (coverage 0.900–0.911, every cell within 4 SE). Changing the elicitation between calibration and deployment is not (Table 1). The failure is not specific to the legend, and its direction depends on how the confidence *level* shifts rather than on how ranks move: rewording reshuffles ranks less than reversal (Figure 4) yet harms

Table 1: Coverage on three frozen models when the elicitation changes between calibration and deployment, on TruthfulQA (817 questions) and POPQA-ELICIT (7,189 questions); 500 random splits, $M = 4$.

		same elicitation	legend reversed at deployment	prompt reworded at deployment
TruthfulQA	Llama-3.1-8B	.906–.911	.896	.966
	Mistral-7B	.905–.907	.565	.623
	Qwen2.5-7B	.907–.908	.952	.972
PopQA	Llama-3.1-8B	.900–.901	.757	.919
	Mistral-7B	.900–.901	.494	.501
	Qwen2.5-7B	.900–.901	.974	.993

coverage about as much, sometimes by under-covering (Mistral) and sometimes by over-covering with near-full sets (Llama, Qwen). These frozen signals are weak (binned Brier ≈ 0.25 ; sets average 1.8 of 2 labels), so the valid sets are close to trivial; the point is validity, not efficiency. On PopQA, where the reports are more informative (AUROC 0.50–0.74 against 0.40–0.60), the breaks are larger: reversal leaves Llama at 0.76 and Mistral at 0.49, and rewording leaves Mistral at 0.50, while Qwen again turns conservative. The licensing condition of the stage criterion is therefore about the whole elicitation convention, not the legend alone.

6.2 When do groups pay?

Granularity: groups against recalibration, multicalibration and a direct predictor (post-hoc re-analysis). Forecasters, all fitted on the same n_{lab} labelled questions per procedure: the base rate K ; R , Platt recalibration of the report per procedure; G_M , model (1) with M farthest-first groups on 16 PCA dimensions of the candidate’s frozen hidden state; G_{cv} , with $M \in \{1, 2, 4, 8, 16\}$ chosen by inner cross-validation; Rand_M , M random groups; Cat , TruthfulQA’s 39 categories; MC , R multicalibrated over groups and report quartiles on a 40% holdout; D , logistic regression on the 16 representation features and the report. Figure 5 shows the paired differences against R .

On TruthfulQA (Figure 1b; per-budget results in Appendix Figure 5) the recalibrated reports score within 0.003 Brier of the base rate. Groups improve on them in 85–96% of splits at 272 labels (-0.005 to -0.010 Brier), but no test in the pre-declared family survives Holm correction (smallest Holm-adjusted $p = 0.53$; smallest raw one-sided $p = 0.044$), random groups and TruthfulQA’s categories do not help, multicalibration never beats recalibration, and the direct predictor D beats R by 0.032–0.036 (intervals exclude zero from 180 labels) and groups by 0.026–0.027; a post-hoc probe on the full hidden state does better still (Brier -0.042 to -0.057 against R), though the same probe is badly miscalibrated on PopQA (Appendix J). Details, the shared-offset equivalence test and the probe-report

PopQA analysis are in Appendix H.

PopQA-Elicit: the same question with ten times the labels. On 7,189 PopQA questions elicited under the same three procedures (100 random thirds; budgets 100 to 2,396; the pre-declared Holm family is G_4-R over three models and five budgets) the reports are more informative (AUROC 0.50–0.74), recalibration beats the base rate by 0.019–0.033 Brier, and the registered question gets a sharper answer (Table 2; Brier differences $\times 10^3$, corrected 95% intervals). One of the fifteen tests in the pre-declared family survives Holm correction (Mistral at 2,396 labels, Holm-adjusted $p = 0.047$); none at 2,000 labels does (smallest adjusted $p = 0.108$). Descriptively, the per-model corrected intervals exclude zero from 2,000 labels for every model, groups then also beat random groups of the same number (-1.7 to -2.2), and random groups cost a little at every budget. As a secondary contrast, groups with M chosen by cross-validation beat recalibration at both large budgets for every model (one-sided $p \leq 0.001$ at 2,396). The best granularity grows with the budget: cross-validation picks 16 groups in 14–21% of splits at 100 labels and in 58–71% at 2,396, and G_{cv} doubles the fixed- M gain. The gain is nevertheless small (under 1% of the recalibrated Brier score with four groups, 1.5–2.1% with M chosen by cross-validation) and the direct predictor’s is 3–15 times larger; it is ahead of groups at every budget from 1,000 labels, with intervals excluding zero for every model from 2,000. The direct predictor’s gain differs by model (-25 , -17 , -7 for forward AUROCs 0.59, 0.50, 0.69) and is smallest for the most informative report, but three models do not establish a trend; the groups’ gain is about the same for all three, as SYNTHETIC-RELIABILITY would lead one to expect. Multicalibration again never beats recalibration, groups again leave conformal sets unchanged, and the direct predictor’s sets are smaller for Llama and Mistral (by 0.12 and 0.09 labels; for Qwen the interval includes zero), with more singleton sets (0.41–0.45 against 0.32–0.40) that are more often correct (0.75–0.78 against 0.66–0.76).

Binned calibration errors and the PopQA-Probe con-

Table 2: POPQA-ELICIT: Brier differences against procedure-specific recalibration ($\times 10^3$; below zero: better), averaged over the three procedures, with Nadeau-Bengio-corrected 95% intervals over 100 random splits.

labels	model	G_4-R	$G_{cv}-R$	$D-R$
300	Llama	-0.6 [-10.4, 9.3]	0.1 [-13.6, 13.7]	-17.3 [-49.7, 15.1]
	Mistral	0.0 [-9.7, 9.8]	-0.1 [-11.5, 11.3]	-8.2 [-24.6, 8.1]
	Qwen	-0.7 [-10.2, 8.8]	-0.8 [-11.8, 10.2]	-2.1 [-13.6, 9.5]
2,396	Llama	-1.7 [-2.9, -0.4]	-3.4 [-5.5, -1.3]	-25.4 [-36.2, -14.6]
	Mistral	-1.5 [-2.6, -0.4]	-3.9 [-5.8, -2.1]	-16.8 [-20.2, -13.4]
	Qwen	-2.0 [-3.8, -0.2]	-4.5 [-6.6, -2.3]	-7.0 [-9.1, -4.9]

found are discussed in Appendix H.

Synthetic-Reliability and Real-Geometry (post hoc). With a latent signal, a group effect of spread τ (32 clusters, or linear in the features) and reports of informativeness κ , on Gaussian features and on the real PopQA hidden states with a known $\mathbb{P}(Y = 1 | x)$, groups lose to recalibration without a group effect at every budget, win only once labels suffice when there is one, and never beat the direct predictor except where there is nothing to find ($\tau = 0$, at least 1,000 labels); κ changes neither comparison (Appendix H).

Conformal sets (post hoc). Every forecaster is valid; groups do not shrink sets (every corrected interval includes zero), the direct predictor’s sets are smaller on PopQA for Llama and Mistral (on TruthfulQA its intervals include zero), and one threshold per group costs efficiency at these budgets (Appendix H). In a post-hoc check that holds the score fixed and changes only the partition, representation cells change set size by at most 0.006 on PopQA and enlarge it on TruthfulQA, where cells fall below the calibration floor; changing the score from the report to the direct predictor shrinks sets by 0.12–0.16 (Appendix J).

6.3 Carrying a calibration across procedures

Transport across procedures (Label-Free-Transfer, Failure-Modes, Controlled-Shifts). A forecaster and its conformal thresholds are fitted under procedure A on disjoint halves of the labelled third and deployed under B. A label-free map T is fitted on the unlabelled third, elicited under both procedures: the quantile map $F_A^{-1} \circ F_B$, or paired isotonic regression of V_A on V_B . The labelled reference recalibrates under B on 120 questions (60 for Platt scaling, 60 for the threshold). Groups do not protect against the change, because their forecast reads the report: under the real rewording, representation groups cover 0.60 for Mistral on TruthfulQA and 0.73 for Qwen on PopQA (report bins: 0.55, 0.92), and only a predictor that ignores the report is invariant (0.90–0.91 under every shift), at a cost in set size once

the report is informative. A monotone map of B’s reports onto A’s scale does carry the calibration on average. Paired isotonic regression restores coverage for a monotone change (trained question-only reporter on TriviaQA, one split: $0.21 \rightarrow 0.910$ against 0.909 under A, set size unchanged; its label-loss mass of 0.013 bounds the loss at 0.025 with probability 0.95); the quantile map recovers most of the loss otherwise without widening sets (Mistral on TruthfulQA: $0.60\text{--}0.63 \rightarrow 0.89\text{--}0.90$); on PopQA no map is reliably exact (0.84–0.95), while 120 labels under the new procedure match the target on average. The repair is marginal, not per group: net of the no-shift baseline, the worst frozen group falls up to 0.18–0.19 short under a map on PopQA, and also 0.08–0.15 short under labelled recalibration on TruthfulQA, which uses one threshold for all groups. Synthetic changes of real reports (CONTROLLED-SHIFTS), the same shifts applied to every method (FAILURE-MODES), oracle, direction and permutation controls, and the label-loss bound with its estimation error are in Appendix I.

6.4 Checks of the theory

Known-Law and Answerability. With a known conditional law the decomposition is exact, cell-averaged error is minimised by one cell (Corollary 5) and the pointwise optimum is interior in all four arms, but the pre-set rate check passed only for k -means cells at $d = 1$ and failed in the other three, including both farthest-first arms (growth is faster; Remark 15, added afterwards). In a synthetic answerability model, calibrating on the answerable set gives the coverage Example 11 predicts (Appendices K.1, K.2).

7 RELATED WORK

Language models can report useful confidence verbally or through a scale (Kadavath et al., 2022; Lin et al., 2022a; Tian et al., 2023; Xiong et al., 2024), and their outputs are sensitive to prompt format, option order and label wording (Zhao et al., 2021; Sclar et al., 2024; Zheng et al., 2024); we measure what that sensitivity does to a calibration carried from one elicitation

to another. Binned recalibration trades bias against variance in the number of bins (Kumar et al., 2019; Gupta and Ramdas, 2021; Sun et al., 2023), and multicalibration asks for calibration on every group of a rich family (Hébert-Johnson et al., 2018; Kim et al., 2019); closest to us, Detommaso et al. (2024) multicalibrate LLM confidence over groups that include prompt-embedding clusters. We ask how finely to group under a label budget, with groups fixed before elicitation. Coverage within frozen cells is the Mondrian guarantee (Vovk et al., 2005; Vovk, 2012), whose Corollary 5 only aggregates the known per-cell Beta law (Ding et al., 2023); the rate behind Theorem 8 is established for localized conformal prediction (Conrad et al., 2026; Min et al., 2026) (the partition-based method of Lei and Wasserman (2014) gives the corresponding efficiency rate), and we claim no new rate; group-conditional coverage without a fixed partition is studied by Gibbs et al. (2023). Conformal methods for language models (Quach et al., 2024; Mohri and Hashimoto, 2024; Cherian et al., 2024) fix the elicitation. Under shift, conformal guarantees are restored by reweighting (Tibshirani et al., 2019; Podkopaev and Ramdas, 2021); selecting on an outcome-defined subset is a calibration/test law mismatch (Wang, 2026). Unlabelled data have carried a calibration to new inputs by quantile matching (Yilmaz and Heckel, 2022), optimal transport of scores (Correia and Louizos, 2025) or a learned map of paired inputs across modalities, with a discrepancy bound that needs a slack assumption (Doula, 2026); here the questions are fixed and only the elicitation changes, so a monotone map acts on the one-dimensional report and the label-loss mass bounds the coverage change directly. Appendix E gives the full discussion.

8 CONCLUSION

The registered question has a conditional answer. Groups learned from a frozen representation add calibration structure beyond recalibrating a confidence report only when reliability differs between groups and labels are plentiful (on PopQA, one pre-declared test at 2,396 labels; the gain is about 1–2% of the Brier score), and even then a direct predictor on the same representation does better; without group structure the cost of estimating groups outweighs their benefit at every budget we tried. Groups do not carry a calibration across procedures, because the report enters their forecast; a map between the procedures’ reports does, exactly when the change is monotone and partly otherwise, and with informative reports no label-free map was reliably exact. In practice, treat a prompt or scale edit as a change of procedure:

- *If the procedure is fixed*, fit a direct predictor on representation and report when hidden states are available; use frozen groups only where per-group coverage guarantees are required, for which Theorem 2 holds under the stage criterion.
- *If the procedure may change*, label about 120 questions under the new procedure and recalibrate: in our runs this matched the target coverage on average and, when the new prompt was the more informative one, gave better forecasts than carrying the old calibration over. Without labels, use a predictor that does not read the report (invariant, at a cost in set size), or map the new reports onto the old scale with unlabelled questions elicited both ways and bound the coverage loss by the label-loss mass plus its Hoeffding term; the repair holds on average, not per group, so per-group guarantees need per-group recalibration. Without hidden states (an API model), only recalibration and the label-free maps are available, since both use the report alone.

Limitations. Groups need white-box hidden states; the models are 7–8B; candidates are fixed rather than the model’s own answers; procedures are two legends and two wordings; groups are capped at 16; the review-driven re-analyses are post hoc, and POPQA-ELICIT was designed after the TruthfulQA result, with its Holm family declared before its run (Appendix D). Learned non-monotone maps and other procedure changes are future work.

AI use statement

In this work, we used generative AI tools to write and test analysis code, to run and summarise experiments, to search and summarise related literature, to draft and revise text, and to draft proofs. Criteria for most experiments were fixed in dated notes before they ran (exceptions are listed in Appendix A), and every result reported was produced by code that we reviewed. All AI-assisted proofs were checked by the authors, and all citations were verified against their sources. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

References

- Abbasi-Yadkori, Y., Kuzborskij, I., Stutz, D., György, A., Fisch, A., Doucet, A., Beloshapka, I., Weng, W.-H., Yang, Y.-Y., Szepesvári, C., Cemgil, A. T., and Tomasev, N. (2024). Mitigating LLM hallucinations via conformal abstention. *arXiv preprint arXiv:2405.01563*.

- Bao, Y., Huo, Y., Ren, H., and Zou, C. (2024). Selective conformal inference with false coverage-statement rate control. *Biometrika*.
- Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2021). The limits of distribution-free conditional predictive inference. *Information and Inference: A Journal of the IMA*, 10(2):455–482.
- Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845.
- Bastani, O., Gupta, V., Jung, C., Noarov, G., Ramalingam, R., and Roth, A. (2022). Practical adversarial multivalid conformal prediction. In *Advances in Neural Information Processing Systems*, volume 35, pages 29362–29373.
- Bates, S., Candès, E., Lei, L., Romano, Y., and Sesia, M. (2023). Testing for outliers with conformal p-values. *The Annals of Statistics*, 51(1):149–178.
- Berthier, L., Shokry, A., Moreaud, M., Ramelet, G., and Dieuleveut, A. (2026). Self-organized conformal prediction: Reducing regional coverage gaps with unsupervised group discovery. *arXiv preprint arXiv:2606.29403*.
- Cherian, J. J., Gibbs, I., and Candès, E. J. (2024). Large language model validity via enhanced conformal prediction methods. In *Advances in Neural Information Processing Systems*, volume 37. arXiv:2406.09714.
- Conrad, A., Isaev, R., Belomestny, D., Moulines, E., and Samsonov, S. (2026). Beyond marginal validity: Finite-sample guarantees for localized conformal prediction. *arXiv preprint arXiv:2608.06206*.
- Correia, A. H. C. and Louizos, C. (2025). Non-exchangeable conformal prediction with optimal transport: Tackling distribution shifts with unlabeled data. In *Advances in Neural Information Processing Systems*, volume 38.
- Dasgupta, S. and Long, P. M. (2005). Performance guarantees for hierarchical clustering. *Journal of Computer and System Sciences*, 70(4):555–569.
- Detommaso, G., Bertran, M. A., Fogliato, R., and Roth, A. (2024). Multicalibration for confidence scoring in LLMs. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 10624–10641. PMLR.
- Ding, T., Angelopoulos, A. N., Bates, S., Jordan, M. I., and Tibshirani, R. J. (2023). Class-conditional conformal prediction with many classes. In *Advances in Neural Information Processing Systems*. arXiv:2306.09335.
- Doula, A. (2026). Conformal calibration transfer. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.
- Dwork, C., Kim, M. P., Reingold, O., Rothblum, G. N., and Yona, G. (2021). Outcome indistinguishability. In *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing*, pages 1095–1108. ACM.
- Futami, F. and Fujisawa, M. (2024). Information-theoretic generalization analysis for expected calibration error. In *Advances in Neural Information Processing Systems*, volume 37. arXiv:2405.15709.
- Gibbs, I., Cherian, J. J., and Candès, E. J. (2023). Conformal prediction with conditional guarantees. *arXiv preprint arXiv:2305.12616*.
- Globus-Harris, I., Harrison, D., Kearns, M., Roth, A., and Sorrell, J. (2023). Multicalibration as boosting for regression. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 11459–11492. PMLR.
- Gonzalez, T. F. (1985). Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306.
- Gopalan, P., Kalai, A. T., Reingold, O., Sharan, V., and Wieder, U. (2022). Omnipredictors. In *13th Innovations in Theoretical Computer Science Conference (ITCS 2022)*, volume 215 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 79:1–79:21. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.
- Guan, L. and Tibshirani, R. (2022). Prediction and outlier detection in classification problems. *Journal of the Royal Statistical Society: Series B*, 84(2):524–546.
- Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR.
- Gupta, C. and Ramdas, A. K. (2021). Distribution-free calibration guarantees for histogram binning without sample splitting. *arXiv preprint arXiv:2105.04656*.
- Györfi, L., Kohler, M., Krzyżak, A., and Walk, H. (2002). *A Distribution-Free Theory of Nonparametric Regression*. Springer Series in Statistics. Springer, New York.
- Hébert-Johnson, Ú., Kim, M. P., Reingold, O., and Rothblum, G. N. (2018). Multicalibration: Calibration for the (computationally-identifiable) masses.

- In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1939–1948. PMLR.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.
- Hore, R. and Barber, R. F. (2025). Conformal prediction with local weights: randomization enables robust guarantees. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):549–578.
- Izbicki, R., Shimizu, G., and Stern, R. B. (2022). CD-split and HPD-split: Efficient conformal regions in high dimensions. *Journal of Machine Learning Research*, 23(87):1–32.
- Jiang, Z., Araki, J., Ding, H., and Neubig, G. (2021). How can we know when language models know? On the calibration of language models for question answering. *Transactions of the Association for Computational Linguistics*, 9:962–977.
- Jin, Y. and Ren, Z. (2024). Confidence on the focal: Conformal prediction with selection-conditional coverage. *arXiv preprint arXiv:2403.03868*.
- Joshi, M., Choi, E., Weld, D. S., and Zettlemoyer, L. (2017). TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611. Association for Computational Linguistics.
- Jung, C., Noarov, G., Ramalingam, R., and Roth, A. (2023). Batch multivalid conformal prediction. In *International Conference on Learning Representations*.
- Kadavath, S. et al. (2022). Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.
- Kapoor, S., Gruver, N., Roberts, M., Collins, K., Pal, A., Bhatt, U., Weller, A., Dooley, S., Goldblum, M., and Wilson, A. G. (2024). Large language models must be taught to know what they don’t know. In *Advances in Neural Information Processing Systems*, volume 37, pages 85932–85972.
- Kim, M. P., Ghorbani, A., and Zou, J. (2019). Multiaccuracy: Black-box post-processing for fairness in classification. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*, pages 247–254. ACM.
- Kiyani, S., Pappas, G. J., and Hassani, H. (2024). Conformal prediction with learned features. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 24749–24769. PMLR.
- Kuhn, L., Gal, Y., and Farquhar, S. (2023). Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations*.
- Kumar, A., Liang, P., and Ma, T. (2019). Verified uncertainty calibration. In *Advances in Neural Information Processing Systems*, volume 32. arXiv:1909.10155.
- Kumar, B., Lu, C., Gupta, G., Palepu, A., Belamy, D., Raskar, R., and Beam, A. (2023). Conformal prediction with large language models for multi-choice question answering. In *ICML 2023 Workshop on Neural Conversational AI (TEACH)*. arXiv:2305.18404.
- Lei, J. and Wasserman, L. (2014). Distribution-free prediction bands for non-parametric regression. *Journal of the Royal Statistical Society: Series B*, 76(1):71–96.
- Lin, S., Hilton, J., and Evans, O. (2022a). Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*.
- Lin, S., Hilton, J., and Evans, O. (2022b). TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252. Association for Computational Linguistics.
- Mallen, A., Asai, A., Zhong, V., Das, R., Khashabi, D., and Hajishirzi, H. (2023). When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822. Association for Computational Linguistics.
- Min, Y., Peng, L., and Zou, C. (2026). A unified theory of conditional coverage in conformal prediction with applications. *arXiv preprint arXiv:2605.11602*.
- Mohri, C. and Hashimoto, T. (2024). Language models with conformal factuality guarantees. *arXiv preprint arXiv:2402.10978*.
- Moreno Cencerrado, I. V., Padrés Masdemont, A., Gonzalez Hawthorne, A., Africa, D. D., and Pachiardi, L. (2026). No answer needed: Predicting LLM answer accuracy from question-only linear probes. <https://github.com/ivanvmoreno/correctness-model-internals>. Code repository, commit aa7270c; venue and year to be confirmed by the author.

- Murphy, A. H. (1973). A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600.
- Nadeau, C. and Bengio, Y. (2003). Inference for the generalization error. *Machine Learning*, 52(3):239–281.
- Pearce, J. (2026). Calibration toolbox: A python library for model calibration evaluation. <https://github.com/Jonathan-Pearce/calibration-toolbox>. Version 0.1.0, commit 1c6f60b.
- Peyré, G. and Cuturi, M. (2019). Computational optimal transport: With applications to data science. *Foundations and Trends in Machine Learning*, 11(5–6):355–607.
- Pezeshkpour, P. and Hruschka, E. (2024). Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2006–2017. Association for Computational Linguistics.
- Platt, J. C. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, Cambridge, MA.
- Podkopaev, A. and Ramdas, A. (2021). Distribution-free uncertainty quantification for classification under label shift. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 844–853. PMLR.
- Quach, V., Fisch, A., Schuster, T., Yala, A., Sohn, J. H., Jaakkola, T. S., and Barzilay, R. (2024). Conformal language modeling. In *International Conference on Learning Representations*. arXiv:2306.10193.
- Redmond, M. and Baveja, A. (2002). A data-driven software tool for enabling cooperative information sharing among police departments. *European Journal of Operational Research*, 141(3):660–678.
- Santambrogio, F. (2015). *Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling*, volume 87 of *Progress in Nonlinear Differential Equations and Their Applications*. Birkhäuser, Cham.
- Sciar, M., Choi, Y., Tsvetkov, Y., and Suhr, A. (2024). Quantifying language models’ sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations*.
- Shabat, E., Cohen, L., and Mansour, Y. (2020). Sample complexity of uniform convergence for multicalibration. In *Advances in Neural Information Processing Systems*, volume 33, pages 13331–13340.
- Shen, M., Das, S., Greenewald, K., Sattigeri, P., Wornell, G. W., and Ghosh, S. (2024). Thermometer: Towards universal calibration for large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 44687–44711. PMLR.
- Sun, Z., Song, D., and Hero, A. (2023). Minimum-risk recalibration of classifiers. In *Advances in Neural Information Processing Systems*, volume 36. arXiv:2305.10886.
- Tian, K., Mitchell, E., Zhou, A., Sharma, A., Rafailov, R., Yao, H., Finn, C., and Manning, C. D. (2023). Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models finetuned with human feedback. In *Proceedings of EMNLP*.
- Tibshirani, R. J., Foygel Barber, R., Candès, E. J., and Ramdas, A. (2019). Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, volume 32.
- Ulmer, D., Gubri, M., Lee, H., Yun, S., and Oh, S. J. (2024). Calibrating large language models using their generations only. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15440–15459. Association for Computational Linguistics.
- Vovk, V. (2012). Conditional validity of inductive conformal predictors. In *Proceedings of the Asian Conference on Machine Learning*, volume 25 of *PMLR*, pages 475–490.
- Vovk, V., Gammerman, A., and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.
- Wang, C. (2026). When does trimming help conformal prediction? a retained-law diagnostic under calibration contamination. *arXiv preprint arXiv:2605.06204*.
- Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., and Hooi, B. (2024). Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations*.
- Yilmaz, F. F. and Heckel, R. (2022). Test-time recalibration of conformal predictors under distribution shift based on unlabeled examples.
- Zadrozny, B. and Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699. ACM.

- Zhao, Z., Wallace, E., Feng, S., Klein, D., and Singh, S. (2021). Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 12697–12706. PMLR.
- Zheng, C., Zhou, H., Meng, F., Zhou, J., and Huang, M. (2024). Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations*.
- Zhou, H., Wan, X., Proleev, L., Mincu, D., Chen, J., Heller, K. A., and Roy, S. (2024). Batch calibration: Rethinking calibration for in-context learning and prompt engineering. In *International Conference on Learning Representations (ICLR)*.

CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. Yes (Sections 2–5).
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. Yes: sample-size dependence in Theorem 8; farthest-first traversal costs $O(Mn_{\text{fit}})$ distance evaluations.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. Yes (supplementary material).
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. Yes.
 - (b) Complete proofs of all theoretical results. Yes (main text and Appendices B–C); Conjecture 16 is stated as a conjecture.
 - (c) Clear explanations of any assumptions. Yes.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). Yes (supplementary material).
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). Yes (Section 6 and the dated protocol notes).
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). Yes.
- (d) A description of the computing infrastructure used. Yes: all analyses run on a single laptop CPU from cached model outputs.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. Yes.
 - (b) The license information of the assets, if applicable. Yes (Communities and Crime: CC BY 4.0; model licences as published).
 - (c) New assets either in the supplemental material or as a URL, if applicable. Not Applicable.
 - (d) Information about consent from data providers/curators. Not Applicable.
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. Not Applicable.
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. Not Applicable.
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. Not Applicable.
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. Not Applicable.

Table 3: Experiment names and the identifiers used in the dated notes and run directories.

experiment	identifier in notes and runs
KNOWN-LAW	X1
COMMUNITIES	X2
ANSWERABILITY	X4
TRAINED-REPORTERS, LEGEND-SWAP, POPQA-PROBE (REPORTERS)	E1, E2, E3
ELICITATION-SHIFT	E2-frozen
GRANULARITY	E5
SYNTHETIC-RELIABILITY	E7
LABEL-FREE-TRANSFER	E8
POPQA-ELICIT	E9
CONTROLLED-SHIFTS	E10
REAL-GEOMETRY	E11
FAILURE-MODES	E12

How Finely to Group? Learned Partitions for Calibrating LLM Confidence across Elicitation Procedures: Supplementary Materials

A PROTOCOL DISCLOSURES

Each experiment’s criteria were written in a dated note before it ran; the notes are in the supplementary material. The notes and run directories use short identifiers: Departures, in order:

- *KNOWN-LAW*. The estimand, functional definitions and agreement criterion were fixed after an earlier KNOWN-LAW run had been seen (not a pre-registration). Amendments before the reported run: the fit of $\hat{a}$ excludes (n_{cal}, M) pairs with full-set test mass. Added after the run, changing no outcome: an INCONCLUSIVE outcome, a floor-ceiling table, a minimum smoothed M^* . The farthest-first arm was added after the cell-diameter lemma was proved; Remark 15 and its $d = 2$ fit are post hoc.
- *ANSWERABILITY*. The per-cell check uses ± 4 seed standard errors instead of a fixed 0.01 (changed after a 4-seed smoke run).
- *REPORTERS and ELICITATION-SHIFT*. Criteria fixed before the runs; the re-randomised-split analyses were defined after the first outcomes and are post hoc.
- *GRANULARITY*. The pre-set analysis (fixed penalty, naive intervals; Appendix K) is superseded by a re-analysis defined after a simulated review: tuned penalties, controls (base rate, random and category groups, holdout multicalibration), Nadeau-Bengio intervals with a Holm-corrected primary family, transfer, and conformal sets at three levels. After smoke tests and before the full runs, the report coefficient was penalised a hundred times less than group and feature coefficients; runs started before this change were discarded.
- *SYNTHETIC-RELIABILITY*. The first synthetic version placed 8 clusters in 8 dimensions, which makes the cluster structure linear; the reported version (32 clusters in 4 dimensions) was defined after seeing the first and also rescaled the linear arm. Both versions found that groups never beat the direct predictor.
- *LABEL-FREE-TRANSFER*. Criteria fixed before the run; the labelled reference arms were corrected after a 3-split smoke run (they had been scored against the wrong thresholds).
- *POPQA-ELICIT, CONTROLLED-SHIFTS, REAL-GEOMETRY, FAILURE-MODES*. Added on the last day after a request for more data and harder tests; each note was written before its run, but all were designed after

the TruthfulQA results were known, in particular after the TruthfulQA family of group-versus-recalibration tests had returned no significant result. POPQA-ELICIT declared its own Holm family (three models $\times$ five budgets; one rejection, Mistral at 2,396 labels, adjusted $p = 0.047$). Treating the two datasets as one registered question tested 27 times, no test survives (smallest Holm-adjusted $p = 0.085$). POPQA-ELICIT re-elicits PopQA under the frozen TruthfulQA templates; before any label was joined its batch size was changed from 32 to 1 because a padding check failed (differences up to 0.016), and a loader error that would have read the TruthfulQA reports was caught (it failed loudly) and fixed before any analysis ran.

- *Calibration metrics.* Calibration errors from calibration-toolbox were added to every forecast evaluation as descriptive metrics after the analyses were defined; no criterion uses them.
- *Figures.* The Figure 2 item was chosen by a stated rule (closest to both population median moves); the transport alignment uses lift rather than the raw plan, changed after the first figure because the raw plan is dominated by letter popularity.

B PROOF OF LEMMA ??

Let $\rho_k = \max_i d(X_i, E_k)$, which is non-increasing, and $r_M = \rho_M$. *Step 1 (separation).* For $i < j \leq M + 1$, $d(e_i, e_j) \geq \rho_{j-1} \geq r_M$, so the balls $B(e_i, r_M/2)$ are disjoint. Each has mass $\geq a \min(r_M/2, r_0)^d$ and the masses sum to at most 1, so $r_M \leq \max(2, \text{diam } S/r_0) (a(M + 1))^{-1/d}$ deterministically. *Step 2 (net).* Let $\delta_n = 2(s \log n / (an))^{1/d}$ and take a maximal $\delta_n/2$ -separated set in S ; it has at most $2^d / (a(\delta_n/2)^d)$ points, and each $\delta_n/2$ -ball around them has mass $\geq s \log n / n$. A union bound gives that every such ball contains a sample point with probability $\geq 1 - 2^d n^{1-s} / (s \log n)$, and then $\sup_{x \in S} d(x, D_{\text{fit}}) \leq \delta_n$. *Step 3 (population covering radius).* $\sup_{x \in S} d(x, E_M) \leq r_M + \delta_n$. *Step 4 (diameter).* A cell lies in the ball of radius $r_M + \delta_n$ around its exemplar, so its diameter is at most $2(r_M + \delta_n)$; $\delta_n \leq a^{-1/d} M^{-1/d}$ iff $M \leq n / (2^d s \log n)$, which gives $K M^{-1/d}$. $\square$

Numerical check ($n_{\text{fit}} = 4000$, 10 seeds, exact clipped cells): the log maximum diameter falls with slope -1.00 ($d = 1$) and -0.54 ($d = 2$) in $\log M$ for farthest-first, with $\hat{K} \approx 1.9$ and 2.3 (the lemma’s constants are conservative by about $3\times$); k -means snapped cells give -0.91 and -0.46 .

C PROOF OF THEOREM ??

Work on the event Ω_n of Lemma 7; off it $R_{\text{pt}} \leq 1$, which gives the last term. *Quantisation.* For x in cell j , $|h(x) - C_j| \leq \sup_{x' \in V_j} \sup_s |F_x(s) - F_{x'}(s)| \leq L \text{diam}_j^\beta \leq L K^\beta M^{-\beta/d}$, uniformly in the realised threshold. *Cell masses.* Exemplars are at least r_M apart (Step 1 of Lemma 7), so each cell contains $B(e_j, r_M/2) \cap S$. *Covering step.* On Ω_n , $S \subset \bigcup_j B(e_j, R_M)$ with $R_M \leq r_M + \delta_n$, and $\text{Vol}(S) \geq 1/\bar{c}$; hence $M \omega_d R_M^d \geq 1/\bar{c}$ and $r_M \geq (\bar{c} \omega_d M)^{-1/d} - \delta_n \geq \frac{1}{2} (\bar{c} \omega_d M)^{-1/d}$ whenever $\delta_n \leq \frac{1}{2} (\bar{c} \omega_d M)^{-1/d}$, which with $\delta_n = 2(s \log n / (an))^{1/d}$ is equivalent to $M \leq c'' n / \log n$, $c'' = c\kappa / (4^d s \bar{c})$. The regularity bound gives $p_j \geq a \min(r_M/2, r_0)^d \geq a (\frac{1}{4} (\bar{c} \omega_d M)^{-1/d})^d = c\kappa / (4^d \bar{c} M) = b/M$, using $r_M/2 \geq \frac{1}{4} (\bar{c} \omega_d M)^{-1/d}$ and the assumption $r_0 \geq \frac{1}{4} (\bar{c} \omega_d M)^{-1/d}$. *Variance.* Conditional on D_{fit} , Corollary 5 gives $\text{Var}(C_j) \leq t(1-t)/(n_j+2) + 1/((n_j+1)(n_j+2))$ above the floor; with $n_j \sim \text{Bin}(n_{\text{cal}}, p_j)$ and $\mathbb{E}[1/(n_j+1)] \leq 1/((n_{\text{cal}}+1)p_j)$, summing $p_j \cdot$ over cells gives $t(1-t)M/(n_{\text{cal}}+1)$ without any balance assumption. *Rounding.* $(k_j/(n_j+1) - t)^2 \leq (n_j+1)^{-2}$ and $\mathbb{E}[1/((n_j+1)(n_j+2))] \leq 1/((n_{\text{cal}}+1)(n_{\text{cal}}+2)p_j^2)$ with $p_j \geq b/M$ give the third term. *Floor.* By a Chernoff bound $\mathbb{P}(n_j < n_{\text{cal}} p_j / 2) \leq e^{-n_{\text{cal}} p_j / 8}$; when $n_{\min} \leq n_{\text{cal}} p_j / 2$ a below-floor cell has probability at most $e^{-bn_{\text{cal}}/(8M)}$ and costs at most α^2 per unit mass. $\square$

D LIMITATIONS IN FULL

Not claimed. Conditional coverage in the strong sense (Barber et al., 2021); anything inside the residual cell; optimality of exemplar selection; anything about the encoder.

White-box hidden states are needed for the groups. The number of groups is capped at 16, which cross-validation reaches in most PopQA splits at the largest budgets, and no experiment has a reliability structure that frozen groups represent exactly, so we do not show where a finer partition would overtake the direct predictor. We do not compare with recalibrating a report averaged over procedures, which needs every procedure at deployment. The

models are 7–8B instruct models; the candidates are fixed and label-blind (817 TruthfulQA questions and 7,189 PopQA questions with constructed distractors) rather than the model’s own answer, so “elicitation procedure” means two legends and two wordings, one of them artificial, with one prompt template carried over unchanged from TruthfulQA to PopQA. Real-data intervals come from overlapping splits of one pool and use a conservative correction; the review-driven re-analyses (tuning, controls, corrected intervals, transfer) are post hoc. We did not evaluate token-probability, $P(\text{True})$, self-consistency or semantic-entropy signals, a learned non-monotone transport map, or larger and API-served models. The finite-sample bound covers farthest-first cells in low dimension ($d \leq 2$ here); at the 16 dimensions of GRANULARITY it is vacuous, and the asymptotic rate itself is known from localized conformal prediction.

E EXTENDED RELATED WORK

Elicited confidence and its sensitivity. Language models can report useful confidence in their own answers (Kadavath et al., 2022; Jiang et al., 2021), either verbally or through a scale (Lin et al., 2022a; Tian et al., 2023; Xiong et al., 2024), and can be trained or post-processed to do so (Kapoor et al., 2024; Ulmer et al., 2024); answer-side signals such as semantic entropy are an alternative (Kuhn et al., 2023). Model outputs are sensitive to prompt format, option order and label wording (Zhao et al., 2021; Sclar et al., 2024; Zheng et al., 2024; Pezeshkpour and Hruschka, 2024). We study what that sensitivity does to a calibration made under one elicitation and deployed under another, and measure it against shuffle and redraw references.

Recalibration, grouping and multicalibration. Platt scaling, isotonic regression and histogram binning map a score to a probability (Platt, 1999; Zadrozny and Elkan, 2002; Guo et al., 2017); the bias–variance trade-off in the number of bins is analysed by Kumar et al. (2019); Gupta and Ramdas (2021); Sun et al. (2023); Futami and Fujisawa (2024), and Proposition 17 restates the decomposition of Murphy (1973) in that setting. Multicalibration and multiaccuracy ask for calibration on every group in a rich family (Hébert-Johnson et al., 2018; Kim et al., 2019; Shabat et al., 2020), connect to outcome indistinguishability and omniprediction (Dwork et al., 2021; Gopalan et al., 2022), and can be achieved by boosting (Globus-Harris et al., 2023). The group forecaster G_M is a structured special case: without a penalty, the score equations of a logistic model with one intercept per group make its mean residual zero on every group of the training sample, so it is multiaccurate on its groups in sample (our tuned fits add a small penalty). Closest to us, Detommaso et al. (2024) multicalibrate LLM confidence scores over groups that include clusters of prompt embeddings. We ask a different question — how finely to group under a label budget, with groups fixed before elicitation and forecasts that depend on the procedure — and find that at our budgets a direct predictor on the same representation does at least as well, and that holdout multicalibration costs more labels than it gains.

Conformal prediction with partitions. Coverage within frozen cells is the Mondrian guarantee (Vovk et al., 2005; Vovk, 2012); exact conditional coverage is impossible in general (Barber et al., 2021), which motivates group-conditional, multivalid and function-class-conditional guarantees (Jung et al., 2023; Bastani et al., 2022; Gibbs et al., 2023). Partitions can be clustered (Ding et al., 2023), learned from features (Kiyani et al., 2024), or derived from the conditional density (Izbicki et al., 2022). The exact Beta law of calibration-conditional cell coverage appears in Vovk (2012) and Ding et al. (2023); Corollary 5 only aggregates it across cells, and on auditing a method on its own partition see Berthier et al. (2026). The bias–variance balance behind Theorem 8, with coverage error $n^{-\beta/(2\beta+d)}$ under Hölder continuity of the conditional score law, is established for kernel-localized conformal prediction (Conrad et al., 2026; Min et al., 2026), analysing the randomized method of Hore and Barber (2025), and goes back to the partition-based method of Lei and Wasserman (2014) and to nonparametric partitioning estimates (Györfi et al., 2002); we give the hard-partition analogue for farthest-first cells (Gonzalez, 1985; Dasgupta and Long, 2005) and claim no new rate.

Conformal prediction for language models. Conformal methods give LLM answers set-valued or abstention guarantees (Kumar et al., 2023; Quach et al., 2024), filter claims by self-evaluation (Mohri and Hashimoto, 2024), guarantee coverage conditional on bins of a data-adaptive confidence level (Cherian et al., 2024), and control hallucination rates through abstention (Abbasi-Yadkori et al., 2024). These works fix the elicitation; none asks at which stage a confidence bin is computable, what happens when the elicitation changes between calibration and deployment, or how many bins a label budget supports.

Table 4: Stages of information in an LLM pipeline, with example stratifiers.

Stage	Known	Example stratifiers
0	prompt	exemplar cells in prompt space; pre-answer confidence V_{pre}
1	prompt + sampled answer	post-answer verbalized confidence V_{post} ; semantic entropy
2	+ correctness	the “answerable” set

Shift, selection and transport. Conformal guarantees can be restored under covariate shift by importance weighting (Tibshirani et al., 2019), under label shift by reweighting (Podkopaev and Ramdas, 2021), and degrade gracefully without exchangeability (Barber et al., 2023). Selection on test features is valid when the same rule is applied to calibration data (Bao et al., 2024; Jin and Ren, 2024), while calibrating on an outcome-defined subset is a calibration/test law mismatch (Wang, 2026); Example 11 is a special case. A change of elicitation is a different kind of shift: the covariates are unchanged and the report function changes. Because a question can be elicited under both procedures without being labelled, the coupling of the two reports is observable, and a one-dimensional transport map (Peyré and Cuturi, 2019; Santambrogio, 2015) can carry one report onto the other; Proposition 4 bounds the resulting change in coverage by a crossing mass that needs no labels. Unlabelled data have been used to carry a conformal calibration to a new input distribution, by matching confidence quantiles (Yilmaz and Heckel, 2022), by bounding and reweighting through optimal transport of scores (Correia and Louizos, 2025), or across sensor modalities by transporting paired inputs (Doula, 2026); label-free prompt recalibration (Zhao et al., 2021; Zhou et al., 2024; Shen et al., 2024) instead debiases or rescales within a prompt or task. What is specific here is the case where the questions are fixed and only the elicitation procedure changes: the calibration is carried by a monotone map of the confidence report itself, and the coverage change is bounded by a label-free crossing mass.

F FURTHER THEORY AND PROOFS

Stages in an LLM pipeline.

Example 9 (Stages in an LLM pipeline).

Confidence bins as exemplar cells.

Remark 10 (Confidence bins are exemplar cells on a scalar). When $\phi(x)$ is a scalar confidence signal V (verbalized or otherwise), exemplar cells are intervals of V with cut points at the midpoints of consecutive exemplars. Arbitrary interval cut points (e.g. quantile bins) are not always midpoint-consistent; they are covered by the general Definition 18, which ANSWERABILITY uses.

Proof sketch of Theorem 2.

Proof sketch. Condition on D_{fit} and on the cell labels of all calibration points and the test point. Since γ is frozen, the scores of the n_j calibration points in cell j together with the test score (given $G_{\text{test}} = j$) are exchangeable; the rank of the test score is uniform on $\{1, \dots, n_j + 1\}$, and the k_j -th order statistic covers it with probability $k_j/(n_j + 1) \geq t$. Below-floor cells return the full set. (Vovk et al., 2005; Vovk, 2012). $\square$

Proof of Proposition 3.

Proof. Apply Theorem 2 with the covariate taken to be $\mathcal{I}_t$; the units $(\mathcal{I}_t, Y)$ inherit exchangeability from W . Selecting units by an $\mathcal{I}_t$ -measurable event preserves exchangeability of the retained units provided the test unit is subjected to the same event. $\square$

Outcome selection and answer-computed confidence.

Example 11 (Outcome selection breaks coverage). A special case of the calibration/test law mismatch of Wang (2026). Let $Y \in \{0, 1\}$ with $\mathbb{P}(Y = 1) = p$ (“answerable”). Suppose $S \mid Y=1 \sim U[0, 1]$ and $S \mid Y=0 \sim U[1, 2]$. Calibrate on the answerable set $\{Y = 1\}$ only and apply the threshold q to the full test stream. As $n \rightarrow \infty$, $q \rightarrow t$ and

$$\mathbb{P}\{S_{\text{test}} \leq q\} \rightarrow pt,$$

a coverage gap of $(1 - p)t$ relative to the target t , which vanishes only when $p = 1$. Coverage on the answerable set itself remains t .

Proof. Conditional coverage on $\{Y = 1\}$ is the standard guarantee. On $\{Y = 0\}$, $S > 1 \geq q$ a.s., so coverage is 0. Mix with weights $p, 1 - p$. $\square$

Remark 12 (Selection is a stage-2 stratifier). The answerable set is $\sigma(\mathcal{I}_2)$ -measurable. Example 11 is the failure of Proposition 3’s selection condition. Label-dependent Mondrian taxonomies remain valid *within* category (Vovk et al., 2005); the failure is that the test category is unavailable at decision time, so a threshold calibrated on one category is applied to a population it does not describe. Selection on test *features* by a fixed, data-independent rule is valid (Bao et al., 2024); data-dependent selection rules need adjustment (Jin and Ren, 2024); selection on the *outcome* is not, which is the distinction the stage index makes explicit.

Remark 13 (Answer-computed confidence: when and what it licenses). (a) V_{post} is $\mathcal{I}_1$ -measurable: it licenses post-generation decisions (show or withhold the sampled answer) but has no value at stage 0, so it cannot gate generation. (b) If the test category is computed from a different decode than the one the score is evaluated on, the pair (category, score) at test no longer has the calibration law; decode greedily or fix seeds. (c) Binning by V_{post} gives coverage conditional on the bin, not conditional on latent answerability $a(X)$: $\mathbb{P}\{Y \in C \mid G = j\} \geq t$ does not imply $\mathbb{P}\{Y \in C \mid G = j, a(X) = a\} \geq t$. Whether verbalized confidence tracks what it names is a separate question that we do not measure here.

Proof of Proposition 4.

Proof. (a) $|\mathbb{E}[\mathbf{1}\{Y \in C_T\} - \mathbf{1}\{Y \in C_A\}]| \leq \mathbb{E}|\mathbf{1}\{Y \in C_T\} - \mathbf{1}\{Y \in C_A\}| \leq \mathbb{P}(C_T \neq C_A)$, and the sets depend on (V_A, V_B, G) only; conditioning on $G = j$ gives the per-group statement. If $Y \in C_A$ and $Y \notin C_T$ then $C_A \not\subseteq C_T$, so $\mathbb{P}(Y \in C_A) - \mathbb{P}(Y \in C_T) \leq \mathbb{P}(Y \in C_A, Y \notin C_T) \leq \mathbb{P}(C_A \not\subseteq C_T)$, conditionally on the calibration data and T ; take expectations for the marginal form. The estimation error is Hoeffding’s inequality for the mean of a $[0, 1]$ indicator over m questions that, given the calibration data and T , are i.i.d. and independent of both. (b) $\mathbb{E}[V_A \mid V_B] = g^{-1}(V_B)$ is monotone, so it is the population isotonic regression; then $T(V_B) = V_A$ and the sets coincide. For increasing g , $F_B = F_A \circ g^{-1}$, so $F_A^{-1}(F_B(g(v))) = v$. (c) In one dimension the comonotone coupling is the unique optimal coupling for a strictly convex cost when the marginals are continuous. $\square$

Residual cell.

Remark 14 (Residual cell). Every query is assigned: cell 0 collects queries outside every exemplar radius fitted on D_{fit} . It is calibrated by the same rule as any other cell, so Theorem 2 covers it; its assignment rate can be controlled with conformal outlier tests (Bates et al., 2023; Guan and Tibshirani, 2022). We make no claim about what its members have in common.

Proof of Corollary 5.

Proof. $C_j = F_j(S_{(k_j)}^{(j)})$ is the k_j -th order statistic of n_j i.i.d. uniforms by the probability integral transform; take mean and variance and use $\mathbb{E}(C_j - t)^2 = \text{Var } C_j + (\mathbb{E}C_j - t)^2$. For the approximation, $\sum_j p_j t(1 - t)/(n_{\text{cal}} p_j) = t(1 - t)M/n_{\text{cal}}$, with no balance assumption. $\square$

Proof of Proposition 6.

Proof. Write $h - t = (h - C_j) + (C_j - t)$ on cell j ; the cross term has zero conditional mean. $\square$

Scope of Lemma 7. Proof in Appendix B. Diameters are of cell $\cap S$. The exemplar sets are nested in M (a traversal prefix) but the cells are not. The lemma does *not* cover k -means exemplars, whose Lloyd fixed points can have cells of diameter $\asymp 1$ for $d \geq 2$; KNOWN-LAW therefore runs a farthest-first arm. Assumption 1 is stated for the covariate the cells are built on; when cells are built on a representation $Z = \phi(X)$, read x as z and d as the dimension of Z . At $d = 16$, as in GRANULARITY, the range $M \leq n/(2^d s \log n)$ admits no $M \geq 2$ for any realistic n_{fit} , so the bound below licenses KNOWN-LAW ($d \leq 2$) only and GRANULARITY is not a test of it.

Proof sketch of Theorem 8.

Proof sketch. On the event of Lemma 7, $|h(x) - C_j| \leq L \text{diam}_j^\beta$ uniformly in the threshold, giving the first term. On the same event the exemplars are at least r_M apart and M balls of radius $r_M + \delta_n$ cover S , so $r_M \geq \frac{1}{2}(\bar{c}\omega_d M)^{-1/d}$ when $M \leq c''n_{\text{fit}}/\log n_{\text{fit}}$; each cell contains $B(e_j, r_M/2) \cap S$, so $p_j \geq b/M$. Condition on D_{fit} and apply Corollary 5 with $n_j \sim \text{Bin}(n_{\text{cal}}, p_j)$ and $\mathbb{E}[1/(n_j + 1)] \leq 1/((n_{\text{cal}} + 1)p_j)$; a Chernoff bound controls below-floor cells. Off the event, $R_{\text{pt}} \leq 1$. Full proof in Appendix C. $\square$

Pre-asymptotic exponents, the interior optimum and the forecast analogue.

Remark 15 (Pre-asymptotic exponents). Write the quantisation term as $Q(M) = \exp f(\log M)$ with local exponent $a(u) = -f'(u)$, and the calibration term as BM/n . The stationarity condition $-Q'(M) = B/n$ gives

$$\frac{d \log M^*}{d \log n} = \frac{1}{a + 1 - a'(u)/a} \Big|_{u=\log M^*},$$

which equals $1/(a + 1)$ only when a is constant. A local exponent that rises towards its asymptote $2\beta/d$ makes finite-grid slopes exceed $1/(a + 1)$.

Conjecture 16 (Interior optimum). *If additionally the quantisation term is bounded below by $cM^{-2\beta/d}$ (non-degenerate heterogeneity), the minimiser of $\mathbb{E}[R_{\text{pt}}]$ is interior and of order $n_{\text{cal}}^{d/(d+2\beta)}$. Without such a lower bound the optimum can be $M = 1$ (e.g. F_x constant in x). The mass bound in the proof of Theorem 8 supplies the geometric half (every cell contains a ball of radius $\gtrsim M^{-1/d}$); the remaining gap is that h is evaluated at the realised threshold.*

Proposition 17 (Forecast analogue). *The Brier decomposition of Murphy (1973) as used for binned recalibration by Sun et al. (2023), who derive $B^* \asymp n^{1/3}$. Let $Y \in \{0, 1\}$, let V be a frozen confidence signal and γ a frozen binning of V , and let $\tilde{p}(V)$ be the calibration-sample rate of Y in cell $\gamma(V)$. With $\eta = \mathbb{P}(Y = 1 \mid V)$ and $\eta_j = \mathbb{P}(Y = 1 \mid \gamma(V) = j)$,*

$$\begin{aligned} \mathbb{E}(\tilde{p} - Y)^2 &= \underbrace{\mathbb{E}[\eta(1 - \eta)]}_{\text{free of } M} + \underbrace{\mathbb{E}(\eta_{\gamma(V)} - \eta)^2}_{\text{quantisation}} \\ &\quad + \underbrace{\sum_j p_j \mathbb{E}(\tilde{p}_j - \eta_j)^2}_{\approx \sum_j \eta_j(1 - \eta_j)/n_{\text{cal}}}. \end{aligned}$$

Proof. Condition on V ; $\tilde{p}$ is a function of the cell and the calibration sample, independent of the test Y . Expand $(\tilde{p} - Y) = (\tilde{p} - \eta_\gamma) + (\eta_\gamma - \eta) + (\eta - Y)$; the cross terms vanish because $\mathbb{E}[Y \mid V] = \eta$ and $\mathbb{E}[\eta \mid \gamma] = \eta_\gamma$. $\square$

For groups built from other features (POPQA-PROBE, GRANULARITY), the same identity holds with V replaced by the features the grouping reads; cells empty in the calibration sample fall back to the pooled rate. We restate it for one purpose. Unlike the conformal pointwise gap, the forecast analogue is observable on real data up to a constant: test Brier differences across M are differences in quantisation plus estimation error. This is what TRAINED-REPORTERS, POPQA-PROBE and GRANULARITY measure.

G GENERAL FROZEN BINNINGS

Definition 18 (Frozen stage- t binning). A *stage- t binning* is a map $\gamma : \mathcal{I}_t \mapsto \{0, 1, \dots, M\}$ fixed using D_{fit} only: after fitting, the cell of a query depends on that query alone. Cell 0 is a residual cell. Instances: exemplar and power cells (Definition 1); intervals of a scalar score V , $\gamma = j$ iff $V \in (b_{j-1}, b_j]$, with cut points fitted on D_{fit} ; any Mondrian taxonomy computable from $\mathcal{I}_t$.

Theorem 2 and Propositions 3–19 use only that γ is frozen and $\mathcal{I}_t$ -measurable, so they hold verbatim for Definition 18. Theorem 8 additionally needs the diameter condition, which is stated for exemplar cells.

Q: “Known for singing, this famous ... Her name is Hillary what?” Candidate: Scott Scale: A = 0% ... L = 100% vs A = 100% ... L = 0%

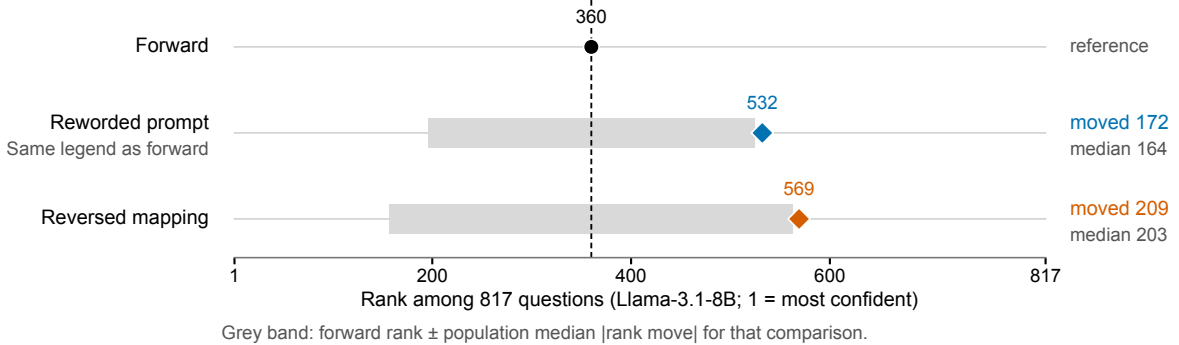


Figure 2: One TruthfulQA item under three elicitations (Llama-3.1-8B). Ranks among all 817 questions (1 = most confident; ties averaged). Grey band: forward rank $\pm$ the population median absolute rank move for that comparison (164 for rewording, 203 for reversal). The item was chosen by a pre-specified rule: closest to both population medians in relative distance, ties to the lowest index.

H FURTHER EXPERIMENTAL RESULTS

Setup in full. TruthfulQA (Lin et al., 2022b) supplies a correct and an incorrect candidate answer for each of its 817 questions; a model is shown the question and one candidate (teacher-forced, no free generation) and asked how confident it is that the candidate is correct, by choosing a letter A–L. The report V is the expected value of the model’s letter distribution decoded through the legend in force. The three procedures are the forward legend, the reversed legend (decoded back to the same 0–100% scale), and a reworded confidence prompt under the forward legend. Figures 2–4 use the correct candidate; ELICITATION-SHIFT and GRANULARITY use one candidate per question chosen by a label-blind hash, so Y is the correctness of that candidate (about half are correct). Labels do not depend on the procedure, so n_{lab} counts labelled questions and every procedure’s forecaster is fitted on the same n_{lab} questions; a further third of the questions, unlabelled, is used only to fit the PCA and the groups. Conformal sets are over $Y \in \{0, 1\}$; a set of size 2 means abstaining. PopQA is held out in two senses: it is a different dataset, and its questions are disjoint from those used to train its probe.

Controlled reporting interventions: three frozen models (Llama-3.1-8B, Mistral-7B, Qwen2.5-7B) on TruthfulQA under three elicitations: forward legend, reversed legend, and a reworded prompt under the forward legend (a fourth, reworded and reversed, is cached but unused). *Held-out factual QA:* PopQA (Mallen et al., 2023) (7,377 questions) with a linear probe on the answer’s hidden state as the report, and TriviaQA (Joshi et al., 2017) answers of Llama-3.1-8B with trained question-only and answer-aware correctness reporters. POPQA-ELICIT repeats the three elicitations, with the TruthfulQA templates unchanged, on all PopQA questions for the three frozen models, pairing each gold answer with a distractor that answers the same relation for a different question (type-consistent in a 256-question audit, where a surface-only classifier could not separate the two: AUROC 0.47); 188 questions whose distractor matches an accepted alias are dropped, leaving 7,189 (budgets up to about 2,400 labels). *Synthetic reliability models:* SYNTHETIC-RELIABILITY. Real-data comparisons use repeated random splits of one pool; because the splits overlap, intervals use the Nadeau–Bengio correction (Nadeau and Bengio, 2003) (variance factor $1/J + n_{\text{test}}/n_{\text{lab}}$), and the comparison that answers the registered question (groups vs. recalibration, three models $\times$ four budgets) is a pre-declared family with Holm correction (Holm, 1979). Every logistic forecaster tunes its penalty by inner cross-validation on the labelled rows; the report’s coefficient is penalised a hundred times less than group or feature coefficients, so tuning does not discard the report.

Ranks. The report is the expected value of the model’s letter distribution, so a same-prompt redraw moves no rank; the references are a reworded prompt and a shuffled-question null. Under reversal a typical question moves 203 [184, 222] (Llama), 240 [221, 260] (Mistral) and 190 [173, 208] (Qwen) positions, against 239 [220, 259] under independent rankings and 139–169 under a reworded prompt; the share of questions that stay in their confidence fifth is 0.240 [0.208, 0.271], 0.202 [0.174, 0.231] and 0.244 [0.217, 0.278], against 0.284–0.337 under

rewording (Figure 4). The reports themselves are weak: AUROC for correctness is 0.54/0.59/0.52 (Llama; forward/reversed/reworded), 0.40/0.46/0.49 (Mistral, whose forward report is anti-informative) and 0.56/0.46/0.60 (Qwen).

Transport plans. Because the forward and reversed prompts are separate calls, the joint law of the two reports for a random question is the average over questions of the product of their letter distributions: an empirical transport plan from forward to reversed reports (Figure 3). Measured on the value scale against two references — an independent redraw under the forward legend (a stable but noisy report) and shuffled questions — the realised cost sits at 85–93% of the way from redraw to shuffle for all three models (reworded prompt: 76–92%); for Mistral under reversal the shuffled and redraw costs differ by only 0.5 points, so its index is not interpretable. For Llama and Mistral the letter plan has almost no lift over independence ($\max |\log_2 \text{lift}|$ of 0.41 and 0.09), so no letter alignment is identified: the reversed report says almost nothing question-specific about the forward one. Qwen shows strong same-letter lift ($\max 5.0$): it partly ignores the legend, which appears on the value scale as a flip. The optimal monotone plan separates what recalibration can repair: most of Qwen’s cost is a level shift ($W_1 = 37$ of 43), most of Llama’s is question-level reshuffling ($W_1 = 7$ of 31) that no recalibration of the report can undo — the case in which structure from a frozen representation (GRANULARITY) has room to help.

Quintile transitions.

Per-split coverage. Per split, the 10th–90th percentiles of coverage are 0.87–0.94 for every same-elicitation arm, 0.50–0.59 and 0.56–0.66 for Mistral under reversal and rewording, and 0.95–0.99 for Qwen under rewording. On TruthfulQA only Mistral under-covers and Llama and Qwen become conservative; on PopQA Llama under-covers too under reversal (0.76), while Qwen again becomes conservative.

Granularity figure.

Granularity on TruthfulQA and PopQA-Probe, in full. On TruthfulQA the reports are nearly uninformative: recalibrated, they score within 0.003 Brier of the base rate. Groups improve on them in 84–95% of splits at 272 labels (−0.005 to −0.010 Brier, a Brier skill over the base rate of 3–5%), but no test in the pre-declared family survives Holm correction (smallest one-sided $p = 0.044$, Qwen at 272 labels), and at 60 labels groups are as likely to hurt as to help. Groups from the representation beat random groups of the same number in 85–96% of splits (intervals include zero); TruthfulQA’s own categories do not help. The number of groups chosen by cross-validation spreads over 4, 8 and 16 at 272 labels, and one group stops being chosen ($P(M = 1)$ falls from 0.23–0.32 at 60 labels to ≤ 0.01): the best granularity grows only in this weak sense. Multicalibration with a holdout never beats the recalibration it starts from: the holdout costs more than the patches gain at these budgets. Group offsets shared across the three procedures differ from per-procedure groups by at most 0.0012 Brier at every budget, but a pre-specified equivalence test with margin ± 0.002 does not establish equivalence (90% corrected intervals at 272 labels, e.g. [−0.0019, 0.0021] for Llama). The direct predictor D, on the same representation, beats R by 0.032–0.036 at 272 labels (Brier skill 14–15%; intervals exclude zero from 180 labels) and beats groups ($M = 4$) by 0.026–0.027; it is ahead of R in 90% of splits even at 60 labels. On PopQA, whose probe report is miscalibrated but discriminative (AUROC 0.73; skill 0.16 after recalibration), groups show no gain at any budget ($|G_4 - R| \leq 0.0005$) while D helps from 1,000 labels (corrected interval; it is ahead in every split from 500).

Binned calibration error. Binned calibration error tells a different and less reliable story: at 272 labels the class-wise calibration error (SCE, calibration-toolbox) of groups and of the direct predictor is higher than recalibration’s (0.066–0.072 and 0.087–0.090 against 0.048–0.054; every corrected interval includes zero), and the constant base rate scores best (0.035). On about 272 test questions with 15 bins, a forecast that varies more spreads over more, sparser bins and pays a larger upward bias, and a constant forecast is calibrated without discriminating; we therefore rank forecasters by Brier score and report binned errors as diagnostics only. The PopQA contrast changes the dataset, the report and the single procedure at once, and its probe is built on the representation the groups use, so it is one observation rather than a de-confounded effect; SYNTHETIC-RELIABILITY isolates the factors.

Synthetic-Reliability: synthetic reliability models (post hoc). A latent signal u and a group effect β determine $\mathbb{P}(Y = 1) = \sigma(\beta + u)$; reports under three procedures are noisy, level-shifted views of u with

informativeness κ ; β varies either across 32 clusters in four dimensions (not representable by a linear function) or linearly in the features, with spread τ . Groups (farthest-first, M by inner cross-validation), recalibration and the direct predictor are fitted on n_{lab} labels (Figure 6; 30 independent seeds per configuration). Without a group effect ($\tau = 0$) groups lose to recalibration at every budget: pure estimation cost. With a group effect they win only once labels suffice (linear structure: $\tau \geq 1$ from 120 labels; cluster structure: $\tau = 2$ at 480 labels). Report informativeness κ barely moves either comparison: the lines for $\kappa \in \{0, 0.5, 1, 2\}$ overlap in every panel. Against the direct predictor groups never win; they tie at best, including when reliability is a non-linear function of the features. A first version placed 8 clusters in 8 dimensions, which is secretly linear; it gave the same answer (Appendix A).

Real-Geometry: known reliability on real representations. The synthetic features above are Gaussian. To keep the geometry real while knowing the truth, we use the 7,377 PopQA hidden states of Llama-3.1-8B as features and simulate only the outcome: $\mathbb{P}(Y = 1) = \sigma(\beta(x) + u)$, with β constant on 32 k -means clusters of a 64-dimensional PCA (which the methods, seeing 16 dimensions, cannot represent exactly) or linear in the features, spread $\tau \in \{0, 0.5, 1, 2\}$, and reports as in SYNTHETIC-RELIABILITY. Because $\eta = \mathbb{P}(Y = 1 \mid x, u)$ is known we score the pointwise error $\mathbb{E}(\hat{f} - \eta)^2$ directly (20 seeds per configuration, budgets 100 to 2,000). The pattern is the synthetic one; differences are in units of 10^{-3} . Without a group effect groups cost +2.5 to +5.9 at 100 labels, shrinking to +0.1 to +0.3 at 2,000. With one they win once labels suffice: cluster structure, $\tau = 1$ from 300 labels (−3 to −10) and $\tau = 2$ from 300 (−17 to −31); linear structure, $\tau \geq 1$ from 100 labels (−6 to −68). The direct predictor is ahead of groups wherever there is structure and at least 300 labels, cluster structure included (+0.3 to +25; the interval excludes zero in 49 of 54 such configurations), and loses to them only when there is nothing to find ($\tau = 0$, $n \geq 1,000$: −0.9 to −1.6, intervals below zero), where cross-validation picks 1.5–3.5 groups on average, close to recalibration, while D pays for 16 null coefficients. Report informativeness again changes nothing.

Conformal sets (post hoc). With forecasters fitted on half of the labels and thresholds on the other half, every forecaster is valid. With one threshold for all questions, the group forecaster’s sets are 0.03–0.07 labels smaller than recalibration’s at $\alpha \in \{0.1, 0.2, 0.3\}$ (e.g. 1.74 against 1.80 for Llama at $\alpha = 0.1$), but every corrected interval includes zero; the direct predictor’s sets are smaller still (1.62–1.65, with 0.35–0.38 singletons against 0.20–0.22). With one threshold per group, the half-split leaves some groups below the calibration floor (15% of test mass receives the full set), so group-Mondrian sets over-cover (0.92) and are no smaller than recalibration’s: per-group guarantees cost efficiency at these budgets. A Brier gain of 0.008 moves few forecasts across a threshold, so the groups’ gain in correctness prediction barely reaches the sets; the direct predictor’s larger gain does. On PopQA all forecasters give sets of the same size.

I TRANSPORT ACROSS PROCEDURES: FULL RESULTS AND CONTROLS

This appendix collects the transport results summarised in Section 6.3. Transport is presented as a repair that is checked empirically: the only guarantee is the label-loss lower bound of Proposition 4(a), and the exact transfer of (b) holds only for the population map under an exactly monotone change.

Label-Free-Transfer: carrying a calibration across procedures without labels. (Criteria fixed in a dated note before the run; the experiment was designed after the coverage breaks of Section 6.1 had been seen and after the reversed trained reporter was known to be anti-correlated with the original, so it is not a test of a prior prediction.) A forecaster and conformal thresholds are fitted under procedure A with labels; deployment is under B. Two label-free maps carry V_B back to A’s scale, both fitted on the unlabelled third of the questions, which are elicited under both procedures: the *quantile map* $F_A^{-1} \circ F_B$, the monotone optimal-transport map between the two marginals (unpaired, increasing), and *paired isotonic* regression of V_A on V_B over the same questions, with its direction (increasing or decreasing) chosen by fit (Proposition 4). Under A, Platt scaling and the conformal thresholds are fitted on disjoint halves of the labelled third (the honest split of Section 2; an earlier run fitted both on the same half and was replaced). The labelled reference recalibrates under B with k labelled questions, half to fit Platt scaling and half to set the threshold. Frozen models: 500 random thirds; trained TriviaQA reporters: one split. Same-procedure coverage is 0.919–0.922 on TruthfulQA, slightly conservative because each threshold now rests on about 34 points per group. Paired isotonic regression repaired the one change that was monotone question by question: it found, without labels, that the reversed question-only reporter runs backwards, and

restored coverage ($0.206 \rightarrow 0.910$, against 0.909 under the original procedure), set size and Brier score (0.1232 against 0.1228; one split), better than recalibrating on 60+60 labels (0.128). The quantile map cannot represent a decreasing change and fails there (0.780). On the frozen models, where questions are reshuffled, paired isotonic regression regresses towards the mean and becomes conservative with near-full sets (0.94–0.99, sizes 1.88–1.99), while the quantile map keeps A’s set size and moves coverage most of the way back (Mistral from 0.60–0.63 to 0.89–0.90). Its near-exact matches for Llama and Qwen come where the forecasts are close to the base rate (Brier ≈ 0.25 , sets ≈ 1.8 of 2 labels), so they say little about informative reports. Mapping within frozen groups changes coverage by at most 0.016. The two-sided crossing mass of Proposition 4(a) is 0.16–0.32 on the frozen models and says little. The one-sided label-loss mass is more useful but must carry its estimation error: for paired isotonic regression it is 0.013 for the question-only reporter, so with $m = 10,647$ test questions and $\delta = 0.05$ (Hoeffding term 0.012) coverage under B is at most 0.025 below A’s, with set sizes unchanged; on Llama and Mistral it is 0.006–0.027, but only because the map widens sets to near-full, and with about 272 test questions per split the Hoeffding term is 0.074. For the quantile map, which keeps set size, it is 0.10–0.17, which certifies little. Full results, including Brier scores, split quantiles, label-loss masses and per-group gaps, are in Appendix K.6.

Transfer on PopQA. On POPQA-ELICIT (100 random thirds of 7,189 questions) the same forecaster, unmapped, falls to 0.53–0.60 under reversal for all three models (and to 0.57 for Mistral under rewording). The quantile map moves it to 0.84–0.95: close for Qwen (0.896, 0.894) and Llama under rewording (0.902), but too high for Llama under reversal (0.951) and too low for Mistral (0.842, 0.874). Paired isotonic regression found that Mistral’s reversal runs backwards (decreasing in every split), but its sets are near-full (0.965 coverage, size 1.92); for Llama it returns the full set, and for Qwen it falls to 0.876. Recalibrating on 60+60 labels gives 0.90 on average everywhere, but judged by the same standard as the maps it is not tighter split by split (10–90% range 0.84–0.95, against 0.88–0.91 for the quantile map on Qwen). Because the reversed legend is more informative than the forward one for Llama and Mistral on PopQA, recalibrating under it also beats carrying the old calibration over (Brier 0.214 and 0.226 against 0.245 and 0.251). With informative reports and a larger pool, then, label-free maps repair most of the break but none is reliably exact. The label-loss mass (Hoeffding term 0.025 at $\delta = 0.05$ for 2,396 test questions) certifies a small loss only where paired isotonic regression widens sets (0.003–0.006 for Llama with near-full sets, 0.06–0.07 for Mistral); for the quantile map it is 0.15–0.26.

Calibration of the carried forecast. Coverage is a property of sets; we also scored the carried forecasts with the class-wise calibration error (SCE, 15 equal-width bins; for two classes it equals the binary ECE of the forecast) from the calibration-toolbox package (Pearce, 2026), pooling test predictions over 50 splits (Figure 7, Appendix K.6). Without a map the forecast is badly miscalibrated (SCE 0.28–0.29 for Mistral, 0.73 for the reversed question-only reporter, against 0.016–0.020 under the original procedure). Paired isotonic regression has the lower SCE on TruthfulQA and TriviaQA (0.035–0.040 for Mistral, 0.016 for the question-only reporter), the quantile map 0.051–0.070 for Mistral and 0.26 for the decreasing change it cannot represent; on PopQA the two are close (0.011 and 0.014 for Mistral under reversal). Binned error is a diagnostic, not a ranking: a constant forecast at the test base rate scores 0.000 in every panel, and for Mistral paired isotonic regression attains its low error by compressing forecasts into $[0.37, 0.61]$, which is also why its sets widen, while the quantile map keeps the spread and the set size. Throughout, we rank forecasters by Brier score and report binned errors only alongside it.

Failure-Modes: what breaks which method. Under one protocol we apply to every method (bins of the report, representation groups with model (1), multicalibration, the direct predictor with and without the report) the shifts that could break it: a monotone level change, a reversal, the real rewording, and a permutation of the reports across a share ρ of questions, each raw, after the quantile map and after paired isotonic regression (100 splits per model; criteria in a dated note). Both pre-set checks pass for all three models: with no shift the maps move coverage by less than 0.005, and after a monotone level change the quantile map restores coverage within 0.01. Three findings. (i) The direct predictor that uses only the representation is invariant to the elicitation by construction and covers 0.904–0.908 under every shift, with sets of 1.61–1.63 against 1.75–1.82 for bins, groups and multicalibration; the direct predictor that also reads the report has sets as small (1.58–1.62) but inherits the report’s shifts. The invariant predictor is not the best calibrated (per-split SCE 0.10–0.11 against 0.05–0.07 for recalibrated bins), so its advantage is in sets and robustness rather than in calibration: when the representation is available, not reading the report is the robust choice. (ii) The real rewording breaks every report-based method for Mistral, groups and multicalibration included (coverage 0.55–0.66); the quantile map restores all of them to

0.87–0.89. Groups do not protect against an elicitation change, because the report enters their forecast. (iii) Paired isotonic regression fails in a direction that depends on the kind of reordering: conservative under additive noise (CONTROLLED-SHIFTS), but under-covering when reports are fully permuted for Llama and Qwen (0.87, 0.86). Report bins also move: under rewording 43–68% of test mass changes bin, so per-bin guarantees attach to different questions than at calibration, while representation groups do not move. On POPQA-ELICIT (50 splits) both pre-set checks pass again and the state-only direct predictor is again invariant (0.899–0.903), but the reports are informative there, so for Qwen its sets are larger than those of report-based methods (1.74 against 1.53–1.62): not reading the report buys robustness at a cost in efficiency once the report carries information. Rewording now hurts representation groups and multicalibration more than report bins (Llama: 0.83–0.84 against 0.91; Qwen: 0.72–0.73 against 0.92), because their forecast reads the report through coefficients fitted under the old wording; the quantile map does not repair them for Llama (0.84) and only partly for Qwen (0.86). Qwen’s forward reports put 63% of their mass beyond 0.02 or 0.98 (15% ties), so its quantile maps are degenerate and fail under reversal (0.55–0.73).

Labelled transfer. Reusing group offsets fitted under one procedure and refitting only the report slope and intercept on k labels under the new one was numerically unstable at small k (split-level Brier scores up to 0.58); we do not draw conclusions from it.

Controlled-Shifts: the transport proposition on real reports. We build procedure B from real procedure-A reports, $\text{logit } V_B = s \text{logit } V_A + b + \sigma \text{sd}(\text{logit } V_A) \varepsilon$, with $s = \pm 1$, $b \in \{0, 1\}$ and question-level noise $\sigma \in [0, 2]$, on the three frozen models (100 splits) and the TriviaQA answer-aware reporter. At $\sigma = 0$ the predictions of Proposition 4 are mathematical, so this part checks the implementation and the finite-sample behaviour of the fitted maps; all three hold on all four datasets, as fixed before the run. Paired isotonic regression recovers A’s coverage for both signs (mean per-split $|\Delta\text{coverage}| \leq 0.0004$), and the quantile map does so only for $s = +1$ (≤ 0.003); under reversal it misses by 0.03–0.07 per split on the frozen models and 0.22 on TriviaQA (mean coverage 0.84–0.90 and 0.69, against 0.90–0.91; for Llama and Qwen the per-split errors largely average out). The crossing mass is near zero at $\sigma = 0$ and grows with σ ; the bound $|\Delta\text{coverage}| \leq \text{crossing mass}$ holds in every split, as it must, because both are computed on the same test questions. The informative part is $\sigma > 0$: at $\sigma = 2$ the quantile map keeps A’s set size and stays at 0.89–0.91 on the frozen models for an increasing change, but falls to 0.86 for Mistral under reversal and to 0.80 and 0.71 for the TriviaQA reporter, while paired isotonic regression regresses towards the mean and becomes conservative (0.97–0.98 with sets of 1.95–1.96 on the frozen models, 0.92 with sets of 1.39 on TriviaQA).

Controls and map variants. These arms were added after an external design review, with criteria fixed in a dated note before the run; they change no number reported above. (i) *Oracles*. In CONTROLLED-SHIFTS the true noiseless inverse of the synthetic change reproduces A’s coverage exactly at $\sigma = 0$ on all four datasets (the identity control is the no-shift arm of FAILURE-MODES, which moves coverage by less than 0.005). Under noise the oracle does worse than the fitted maps ($|\Delta\text{coverage}|$ 0.06–0.08 at $\sigma = 1$, against 0.02 for the quantile map under an increasing change on the frozen models), because noise changes the law of V_B : the useful target is $\mathbb{E}[V_A | V_B]$, which isotonic regression estimates, not g^{-1} . (ii) *Direction*. An isotonic map forced in the wrong direction collapses to a constant forecast: for the reversed question-only reporter the increasing fit gives coverage 0.81 with singleton sets and the decreasing fit 0.910, and in CONTROLLED-SHIFTS a wrong-direction fit misses by 0.09–0.10 at $\sigma = 0$. On the frozen models both directions give near-full sets under the real reversal (coverage 0.94–1.00, sizes 1.88–2.00), so reversing the legend does not flip the decoded report monotonically and neither fixed direction repairs it; on PopQA a decreasing fit gives 0.68 for Qwen. (iii) *Mid-rank map*. A quantile map whose CDFs use mid-ranks, both stored from the transport fold and frozen, agrees with the map used above within 0.001 coverage everywhere, including the TriviaQA reporters, whose reports have 5–32% ties, and Qwen’s forward PopQA report with 15% ties; with atoms no deterministic map matches the two marginals exactly. (iv) *Permutation*. Permuting reports across a share ρ of questions keeps the marginal, so the quantile map cannot see it. At $\rho = 1$ paired isotonic regression ranges from 0.79 to 1.00 across the three models and two datasets: it under-covers for Llama and Qwen on TruthfulQA (0.87, 0.86) and for Mistral on PopQA (0.79), and over-covers in the other three (FAILURE-MODES). (v) *Per-group coverage*. The worst of the four frozen groups is below the target even without a shift (the A→A arm’s worst group is 0.02–0.04 short), so gaps are reported net of that baseline (Appendix K.6). Net, no arm keeps every group: recalibrating on 60+60 labels, which uses one threshold for all groups, leaves the worst group 0.08–0.15 short on TruthfulQA and up to 0.06

Table 5: Post-hoc partition-efficiency check: one correctness model per split is held fixed (R or D) and only the partition changes; mean set size, fallback mass (test mass in cells below the floor) and worst-cell coverage, ranges over the three models, 100 splits.

data (calibration points)	partition	set size (R)	set size (D)	fallback mass	worst cell coverage (R)
TruthfulQA (136)	one cell	1.78–1.80	1.62–1.64	0.00	0.90–0.91
	report bins, $M = 4 / 8$	1.80–1.81 / 1.94–1.95	1.64–1.67 / 1.89–1.90	0.01 / 0.68–0.69	0.85 / 0.87–0.88
	representation cells, $M = 4 / 8$	1.80–1.83 / 1.85–1.89	1.69–1.73 / 1.77–1.83	0.15 / 0.35–0.41	0.88–0.90 / 0.88–0.89
PopQA (1,198)	one cell	1.60–1.68	1.55–1.59	0.00	0.90
	report bins, $M = 4 / 8$	1.61–1.69 / 1.64–1.71	1.56–1.59 / 1.56–1.60	0.00	0.88 / 0.86
	representation cells, $M = 4 / 8$	1.59–1.68 / 1.59–1.68	1.55–1.59 / 1.56–1.59	0.00	0.87–0.88 / 0.85

Table 6: Post-hoc correctness-probe baselines: Brier difference against recalibrating the report ($\times 10^3$; below zero: better) at the largest budget, with Nadeau–Bengio-corrected 95% intervals for P-full.

data	model	P-full	P16	V+P	G ₄	D
TruthfulQA, 272	Llama	−44.7 [−83.3, −6.2]	−30.2	−32.7	−8.8	−34.2
	Mistral	−42.1 [−75.8, −8.3]	−31.2	−31.0	−5.2	−32.1
	Qwen	−57.1 [−95.4, −18.8]	−31.4	−33.6	−9.8	−36.3
PopQA, 2,396	Llama	+74.2 [56.8, 91.5]	−16.8	−23.6	−1.7	−25.4
	Mistral	+69.3 [53.3, 85.3]	+1.9	−12.6	−1.5	−16.8
	Qwen	+127.0 [109.7, 144.2]	+25.7	−0.8	−2.0	−7.0

short on PopQA; the quantile map leaves it up to 0.05 short on TruthfulQA and up to 0.18 short on PopQA; and paired isotonic regression keeps every group only where it widens sets to near-full, falling 0.19 short for Qwen on PopQA. Per-group guarantees after a change of procedure therefore need per-group recalibration under the new procedure, which our label budgets do not support.

J POST-HOC ANALYSES ADDED AFTER THE EXPERIMENT LOCK

These two analyses were added on 5 October, after the experiments had been locked, following an external suggestion; their design was fixed in a dated note before they ran, they reuse the cached states and splits above, and they are descriptive and outside the pre-declared Holm family.

Partition efficiency with a fixed score. Each split fits one correctness model on half of the labelled third (Platt on the report, R, or the direct predictor, D) and holds it fixed; only the partition and its per-cell thresholds, set on the other half with a floor of 20, change (100 splits, averaged over procedures). Coverage is at least 0.90 for every arm. On PopQA, representation cells change the mean set size by at most 0.006 against one cell; report bins enlarge it slightly. On TruthfulQA, every partition finer than one cell enlarges sets, because cells fall below the floor and return the full set. The score matters far more than the partition: replacing R by D shrinks sets by 0.12 (PopQA, Llama) and 0.16 (TruthfulQA). Worst-cell coverage falls with M because the minimum over more, smaller cells is noisier; it is not a per-cell guarantee failure. At these budgets, then, frozen cells do not make conformal sets more efficient for a fixed score.

Correctness-probe baselines. On the same splits and label budgets as GRANULARITY, we compare recalibrating the report (R) with a probe on the full candidate-conditioned hidden state, specified as in Moreno Cencerado et al. (2026) (standardised features, logistic regression with default penalty and balanced class weights; P-full), a tuned, unweighted probe on 16 PCA dimensions (P16), the report combined with a cross-fitted P16 score (V+P), groups (G₄) and the direct predictor (D). Labels spent on a probe count against the same budget. Brier differences against R at the largest budget ($\times 10^3$, corrected 95% intervals): On TruthfulQA the full-state probe is the best forecaster (AUROC 0.81–0.83, against 0.70–0.72 for D), so the 16-dimensional direct predictor of the main text understates how much a learned predictor on the representation can gain. On PopQA the same probe is badly miscalibrated (Brier 0.30–0.34): balanced class weights and an untuned penalty on 4,096 dimensions overfit even at 2,396 labels. The probe alone on 16 dimensions is mixed; combined with the report it is close to D. In every setting, groups gain less than every representation-based predictor except the miscalibrated full probe on PopQA.

Table 7: X4: coverage on the full test stream by stratifier and answerability rate p (50 seeds; worst bin is the seed mean of the worst bin).

p	0.24	0.39	0.50	0.67	0.85	0.99
marginal (stage 0)	.901	.900	.900	.899	.900	.901
V_{pre} bins (stage 0)	.902	.901	.900	.901	.899	.902
V_{post} bins, same decode (stage 1)	.902	.900	.900	.903	.900	.901
V'_{post} bins, fresh decode throughout	.903	.902	.901	.901	.900	.902
calibrate V_{post} , assign test by V'_{post}	.806	.760	.795	.794	.828	.901
worst bin (mean over seeds)	.361	.437	.545	.599	.745	.889
answerable set only (stage 2)	.213	.348	.449	.601	.768	.891

K ADDITIONAL RESULTS

K.1 Known-Law: known conditional law

Known-Law: known conditional CDF. $X \sim \text{U}[0, 1]^d$, $S = (0.4 + 1.6 \bar{X})U$, so F_x is known and R_{pt} , its decomposition and all cell functionals are computed exactly on 12,000 test points. k -means exemplars snapped to D_{fit} ($n_{\text{fit}} = 4000$), $\alpha = 0.1$, floor 20, 19 values of M from 1 to 190 (ratio ≈ 1.3), 8 values of n_{cal} from 300 to 8000, 20 seeds, 1000 whole-seed bootstrap resamples. M^* is the vertex of a local quadratic in $\log M$.

Results, $d = 1$, k -means arm. The decomposition is exact (residual $\leq 10^{-12}$). The cell-level term matches the Beta expectation of Corollary 5 (seed-mean difference 1.4×10^{-5} , SE 2.9×10^{-5}), and cell MSCE is minimised at $M = 1$ for every n_{cal} , as are concentrated undercoverage under the efficiency constraint. The pointwise gap has an interior minimiser at every n_{cal} (99.8% of bootstrap replicates), with M^* rising from 8.4 to 35.2. Its log-log slope is 0.417 [0.395, 0.441]. The quantisation term decays as $M^{-\hat{a}}$ with $\hat{a} = 1.82$ [1.79, 1.84], predicting a slope $1/(\hat{a} + 1) = 0.355$; the difference has CI [0.042, 0.086], inside the pre-set ± 0.10 margin. Descriptively, the slope interval excludes the asymptotic $1/3$ and $\hat{a}$ is below 2: the grid is pre-asymptotic. The floor is inactive at the discrete M^* for every n_{cal} , and $R_{\text{pt}}(M^*)$ is at most 0.47 of the floor ceiling $(1 - t)^2$; dropping $n_{\text{cal}} = 300$, where floor activity begins one grid step past M^* , leaves the result inside the margin (difference CI $[-0.001, 0.061]$; post hoc).

Farthest-first arm (the arm Lemma 7 covers). Interior optimum (rate 1.0), B0 and the decomposition hold at $d = 1$ and $d = 2$; M^* rises from 9.1 to 37.2 ($d = 1$) and from 10.3 to 87.2 ($d = 2$). The mechanism check fails narrowly at $d = 1$ (difference CI [0.050, 0.102] against the ± 0.10 margin) and clearly at $d = 2$ ([0.071, 0.133]); in both the slope exceeds $1/(\hat{a} + 1)$, which Remark 15 explains post hoc. The k -means results below are a secondary arm, not covered by the lemma.

Results, $d = 2$, k -means (gate FAIL). Interior optimum, B0 and the decomposition checks all hold (M^* from 10.0 to 83.1). The slope is 0.654 [0.626, 0.685] against a predicted 0.565 ($\hat{a} = 0.77$); the difference CI [0.069, 0.122] exceeds the margin. The equivalence interval is narrow (width 0.053), so this is a miss, not a lack of power. Two post-hoc diagnostics locate it. The calibration term behaves as assumed: regressing on $\log M$ and $\log n_{\text{cal}}$ gives exponents 0.94 and -0.96 . The floor is active at M^* for $n_{\text{cal}} \leq 800$, but dropping those points does not restore agreement (difference CI [0.007, 0.147]). What fails is the pooled power law for the quantisation term: its local exponent rises from 0.61 ($M \approx 10$) to 1.0 ($M \approx 130$), and Remark 15 shows a rising exponent steepens the M^* slope. A post-hoc fit of that one-parameter correction ($a \approx 0.75$, $a' \approx 0.17$, read off the same curves) gives ≈ 0.65 against the observed 0.654; it is an explanation, not a test.

K.2 Answerability: which stage licenses which bins

Answerability: which stage licenses which bins. DGP of Section 3: latent answerability $a(X)$, two independent decodes, the shown answer's score $S = YU + (1 - Y)(1 + U)$; V_{pre} , V_{post} (same decode as the score) and V'_{post} (a fresh decode) are each cut into five D_{fit} -quantile bins. $n_{\text{fit}} = n_{\text{cal}} = 2000$, $n_{\text{test}} = 20,000$, 50 seeds, six values of $p = \mathbb{P}(Y = 1)$ from 0.24 to 0.99.

Marginal coverage on the full test stream. Every bin of the four legitimate arms has seed-mean coverage ≥ 0.897 (within 4 SE of 0.9). Calibrating on the answerable set gives $p \cdot \mathbb{P}(\text{cover} \mid Y=1)$, matching Example 11 to within 0.001. Using a fresh decode *consistently* is valid; mixing decodes between calibration and test is not, which is

Table 8: X2: Communities and Crime, exemplar cells against one global cell and state strata (8 seeds).

arm	coverage	cell MSCE	cells below floor	full-set test mass	width (finite)
global ($M=1$)	.903	.0003	0	0	.497
exemplar $M=2$	.902	.0005	0	0	.466
exemplar $M=4$	.904	.0011	0.6	.014	.465
exemplar $M=8$	.903	.0017	1.3	.017	.462
exemplar $M=16$	.917	.0034	4.5	.054	.461
exemplar $M=32$	.925	.0055	15.8	.146	.453
state (≈ 44 cells)	.91–.96	.005–.007	≈ 33	$\approx .36$	.44–.57

Table 9: E1–E3: binned recalibration on real data (pre-set split).

signal	stage	Brier $M=1 \rightarrow \min$	M^* (n_{cal} small $\rightarrow$ large)	slope [95%]	status
TRAINED-REPORTERS question-only reporter	0	.155 $\rightarrow$.123	3.2 $\rightarrow$ 15.9	.44 [.41,.47]	FAIL (B0 analogue)
TRAINED-REPORTERS answer-aware reporter	1	.155 $\rightarrow$.105	4.5 $\rightarrow$ 22.7	.47 [.42,.51]	FAIL (B0 analogue)
POPQA-PROBE probe cells	1	.250 $\rightarrow$.209	4.1 $\rightarrow$ 7.6	.15 [.10,.18]	PASS
POPQA-PROBE representation cells	1	.250 $\rightarrow$.243	grid top	—	INCONCLUSIVE

Remark 13(b). No below-floor cells in any arm.

K.3 Communities: Communities and Crime

Communities: Communities and Crime. UCI Communities and Crime (Redmond and Baveja, 2002) ($n = 1994$, 122 predictors, ridge score $|y - \hat{y}|$, 30/40/30 split, 8 seeds). Arms: global, state (the given-group comparator), exemplar cells at $M \in \{2, 4, 8, 16, 32\}$ in the standardised feature space. The pointwise gap is not observable on real data, so only cell-level quantities are reported.

Exemplar cells keep marginal coverage and narrow the finite sets slightly; cell MSCE rises with M , as Corollary 5 predicts. State stratification sends about 36% of test mass to the full set because most states fall below the floor. The worst-state undercoverage is about 0.4–0.5 in every arm with seed SD around 0.2, driven by states with a handful of test rows; no arm is distinguishable on it.

K.4 Trained reporters and probes (Reporters)

Reporters: real data. Criteria fixed in a second dated note before the runs (supplementary material); statuses as pre-set, explanations post hoc. *TRAINED-REPORTERS*: TriviaQA answers of Llama-3.1-8B (53,327 questions; fit/cal/test 26,737/10,597/10,647) with two Qwen3-8B correctness reporters: one sees the question only (stage 0), one also sees the answer (stage 1). *LEGEND-SWAP*: the same reporters under a reversed label legend. *POPQA-PROBE*: PopQA, Llama-3.1-8B hidden states with a linear probe (2,657/1,770/2,950). Here the cells are k -means intervals of the reporter’s logit, a one-dimensional feature that moves with the legend, not farthest-first cells of a frozen representation; Theorem 8 and Lemma 7 are stated for the latter and apply to these cells only through the general binnings of Appendix G. The TriviaQA answers are the model’s own generations (one per question, 80% correct), so the question-only reporter is not facing a balanced pair by construction.

An interior optimum that grows with n_{cal} appears for every signal with enough calibration data. The TRAINED-REPORTERS failures are the cell-level analogue of B0 at the largest n_{cal} only: a fixed 0.005 accuracy difference between the calibration pool and test ($0.005^2 \approx 2.6 \times 10^{-5}$ of a 4.1×10^{-5} total) survives averaging because subsamples overlap. Representation cells in 32 PCA dimensions need more calibration data than PopQA has.

TRAINED-REPORTERS conformal: PASS. Stage-0 and stage-1 cells are valid; stratifying on correctness drops coverage to 0.731. POPQA-PROBE conformal: FAIL on the stage-2 condition, because the probe barely separates correct from incorrect answers and the gap in Example 11 scales with that separation. LEGEND-SWAP: FAIL on the consistent arms (reversed $\rightarrow$ reversed undercovers by 0.004 and 0.009, about 1 and 2.2 SD once calibration variability is counted, which the fixed criterion ignored); the convention change breaks coverage by a wide margin.

Sensitivity: re-randomised splits (post hoc). The two TRAINED-REPORTERS/LEGEND-SWAP failures above trace to a single fixed calibration/test split. A dated addendum (written after those outcomes, before

Table 10: E1–E3 and E2: conformal coverage by stratifier stage ($\alpha = 0.1$, $M = 8$).

conformal arm ($\alpha = 0.1$, $M = 8$)	TRAINED-REPORTERS coverage	TRAINED-REPORTERS worst cell	LEGEND-SWAP / POPQA-PROBE
marginal	.905	—	POPQA-PROBE: .911
question-only cells (stage 0)	.904	.885	
answer-aware cells (stage 1)	.905	.881	POPQA-PROBE rep. cells: .906
answerable set only (stage 2)	.731	—	POPQA-PROBE: .897
legend normal→normal (q / qa)			LEGEND-SWAP: .900 / .905
legend reversed→reversed (q / qa)			LEGEND-SWAP: .896 / .891
calibrate normal, deploy reversed (q / qa)			LEGEND-SWAP: .247 / .647

Table 11: Sensitivity analyses with re-randomised calibration/test splits (post hoc).

analysis	result	key numbers	status
TRAINED-REPORTERS answer-aware reporter	interior M^* , B0 analogue holds	M^* 4.3→24.6, slope .50 [.46,.54]	PASS
TRAINED-REPORTERS question-only reporter	B0 analogue holds; flat minimum	M^* 3.2→12.2, slope .43 [.39,.46]	FAIL (interior .35)
LEGEND-SWAP consistent legends	valid in expectation	coverage .900–.902, worst cell $\geq .898$	PASS
LEGEND-SWAP calibrate normal, deploy reversed	coverage breaks	.241 (q) / .642 (qa)	(part of PASS)
POPQA-PROBE probe cells	interior M^* ; flat minimum	M^* 4.0→13.5, slope .58 [.37,.62]	FAIL (interior .42)
POPQA-PROBE representation cells (PCA-8/32)	still improving at $M = 67$	Brier .250→.246 / .241	INCONCLUSIVE
POPQA-PROBE prompt-only cells (stage 0)	no signal	Brier .2499 at every M	FAIL
POPQA-PROBE conformal	valid arms hold; selection biases by .013	answerable only .887 (11 SE below t)	FAIL (margin .02)

these runs) re-splits the pooled calibration and test rows at random per seed, and checks validity in expectation over calibration draws (the ANSWERABILITY rule). Both interior-criterion failures come from one n_{cal} each, where the Brier curve is flat to within 2×10^{-4} across a factor of two in M and the local quadratic is non-convex. A discrete-argmin check (a dated addendum) puts the minimiser strictly inside the grid in all 1000 bootstrap replicates for KNOWN-LAW ($d = 1, 2$), TRAINED-REPORTERS (both reporters) and POPQA-PROBE probe cells. M^* is weakly identified there, not absent.

K.5 Forecasters and hyperparameters

K.6 Full Label-Free-Transfer results

K.7 Pre-set Granularity analysis

The analysis below used a fixed penalty ($C = 1$), no report up-weighting and naive intervals; it was superseded after review by the tuned, controlled and corrected analysis of Section 6.2, whose conclusions differ: the direct predictor is not the riskiest choice at small budgets, and the groups-versus-recalibration contrasts do not survive correction.

Pre-set Granularity analysis (superseded). (Criteria fixed in a dated protocol note before the run; supplementary material.) *Data.* Controlled reporting interventions: the three frozen models on 817 TruthfulQA questions, one label-blind candidate each, under three procedures (forward legend, reversed legend, reworded prompt). Held-out factual QA: PopQA (7,377 questions) with a linear probe as the confidence report. *Groups:* farthest-first cells on 16 PCA dimensions of the candidate’s frozen hidden state, computed before any confidence prompt. *Forecasters*, all fitted on the same n_{lab} labels: R, Platt recalibration of the report per procedure (= one group); G_M , the same with M group intercepts per procedure; GS_M , group intercepts shared across procedures; MC, R multicalibrated over 8 groups $\times$ 4 report quartiles; D, logistic regression on the 16 representation features and the report. 200 random splits (TruthfulQA), 100 (PopQA); paired differences against R.

These intervals are naive (they ignore the overlap between splits) and the forecasters used a fixed penalty; the corrected analysis of Section 6.2 supersedes every interpretive statement made from this table.

K.8 Lossless coarsening

Proposition 19 (Lossless coarsening). *Let cells j, k have continuous, strictly increasing within-cell score CDFs F_j, F_k with population t -quantiles q_j, q_k and masses $p_j, p_k > 0$. Merging them leaves both children at exact coverage t iff $q_j = q_k$.*

Proof. The merged quantile q_m solves $p_j F_j(q_m) + p_k F_k(q_m) = (p_j + p_k)t$. If $q_j = q_k$ this is solved by $q_m = q_j$ and

Table 12: Forecasters and hyperparameters (GRANULARITY).

forecaster	model	tuning
K	base rate of the labelled questions	none
R	logistic on logit V (Platt)	$C \in \{0.01, 0.1, 1, 10, 100\}$, 5-fold inner CV log loss
G_M	logistic on [logit V , one-hot group], $M \in \{2, 4, 8, 16\}$	as R; report column weighted $\times 10$
G_{cv}	G_M with $M \in \{1, 2, 4, 8, 16\}$ and C chosen jointly	inner CV
Rand_M , Cat	as G_M with random groups or 39 categories	as G_M
D	logistic on [logit V , 16 PCA features]	as G_M
MC	R fitted on 60% of labels; patches on 40% over groups $\cup$ report quartiles	$\tau = 0.01$, min. group 10, ≤ 20 passes
groups	farthest-first exemplars on PCA-16 of the L2-normalised hidden state, fitted on an unlabelled third	–
conformal	split conformal, score $1 - \hat{p}(y)$, floor 20 per cell	–

both children have coverage t . If $q_j < q_k$, then $F_j(q_m) > t > F_k(q_m)$ for the unique solution, which lies strictly between, so one child over-covers and the other under-covers. $\square$

K.9 Top- k retrieval and balanced cells

With $k > 1$ the calibration pool is the union of the calibration rows whose k nearest cells meet the query’s k nearest (each row pooled once). Pools overlap and are query-dependent, so the Mondrian proof of Theorem 2 does not transfer; $k = 1$ recovers it. Cell MSCE remains computable; the exact Beta formula of Corollary 5 does not.

On the TRAINED-REPORTERS data (answer-aware reporter, $M = 16$, $n_{\text{cal}} = 8000$, 50 random re-splits; descriptive, criteria in a dated addendum): $k = 1$ covers 0.900 (worst pool 0.896). For $k = 2$ and $k = 4$ the average rises to 0.947 and 0.935, while the worst query neighbourhood falls to 0.848 and 0.766 (each with about 1,150 test rows). Overlapping pools over-cover on average and under-cover locally; averages hide exactly the failure the partition guarantee excludes.

Balanced cells. Power-diagram cells with D_{fit} -fitted weights reach exactly equal fit masses but do not help. Against Voronoi cells at matched M on the TRAINED-REPORTERS data, the binned forecast’s Brier score is worse at $M = 8$ (paired difference +0.0012 [0.0011, 0.0013]) and unchanged at $M = 16, 32$; worst-cell coverage differences have CIs containing 0. We do not claim a benefit from balancing.

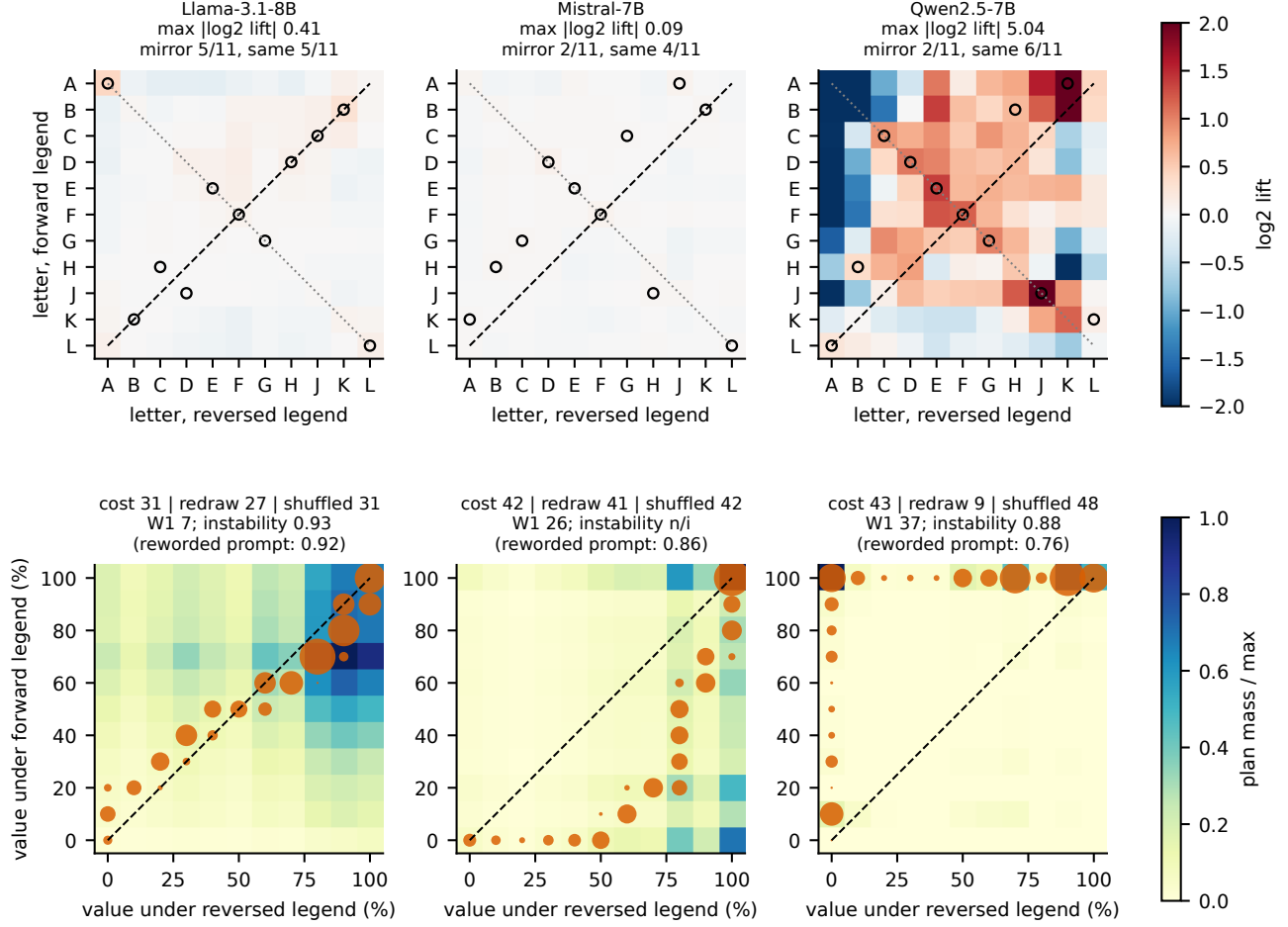


Figure 3: Forward-to-reversed transport plans (817 TruthfulQA questions, both candidates; letters A–L skip I). Top: $\log_2$ lift of letter co-occurrence over independence; circles: Hungarian alignment on lift; dashed: *mirror* (the letter with the same value under the reversed legend, i.e. the legend is read), dotted: *same letter* (the legend is ignored). Bottom: realised plan on the value scale; orange: optimal monotone (W_1) plan between the two marginals, i.e. what recalibrating the report could achieve. Titles: realised cost against an independent redraw under the forward legend (a stable but noisy report) and against shuffled questions; instability = (realised – redraw)/(shuffled – redraw).

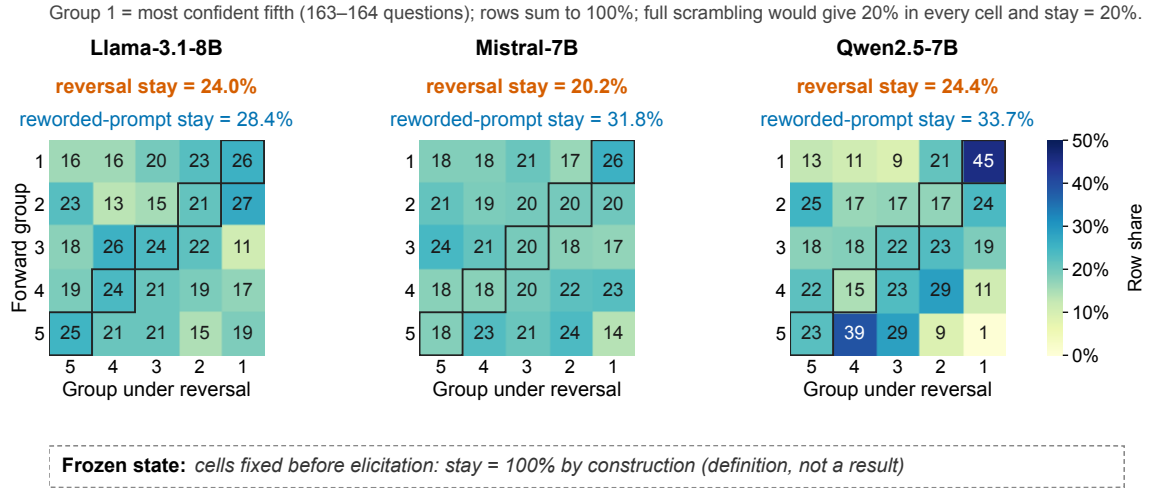


Figure 4: Where each fifth of questions lands under legend reversal (rows: forward fifth, 1 = most confident; columns: fifth under reversal; rows sum to 100%; full scrambling gives 20% everywhere). Stay share (mean diagonal) under reversal is near the 20% null for all three models and below the stay share under prompt rewording. Qwen partly preserves its extremes (45% of its most confident fifth stays there). A binning fixed before elicitation stays put by construction; that is a definition, not a result. What matters for coverage is Section 6 (ELICITATION-SHIFT).

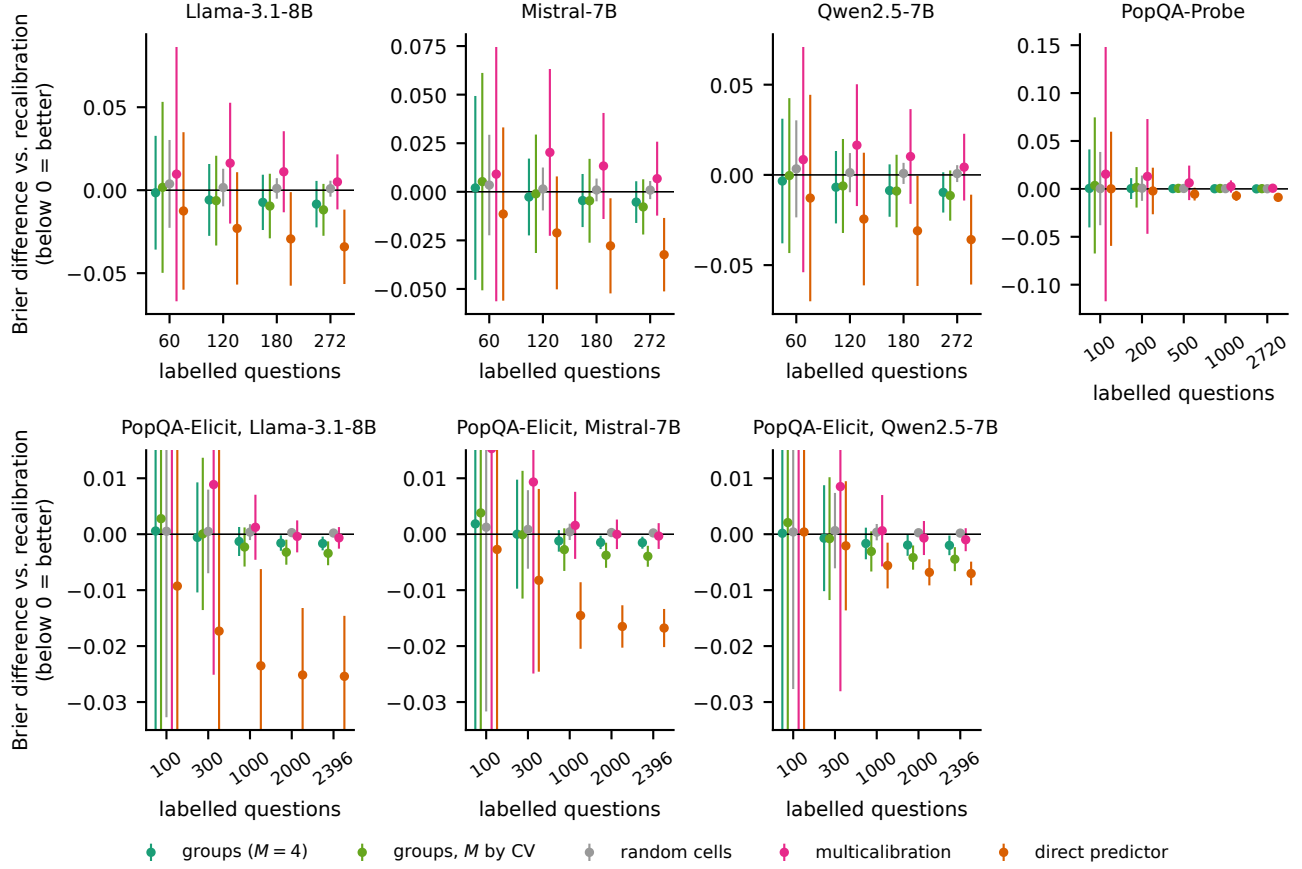


Figure 5: Brier difference against procedure-specific recalibration R (below zero: better), averaged over procedures, with Nadeau-Bengio-corrected 95% intervals. Top: TruthfulQA (three models, 200 random splits) and POPQA-PROBE (probe report, one procedure, 100 splits). Bottom: POPQA-ELICIT (three models, three procedures, 100 splits); the vertical axis is zoomed, so intervals at 100–300 labels run beyond it. Groups: farthest-first cells on 16 PCA dimensions of the frozen hidden state.

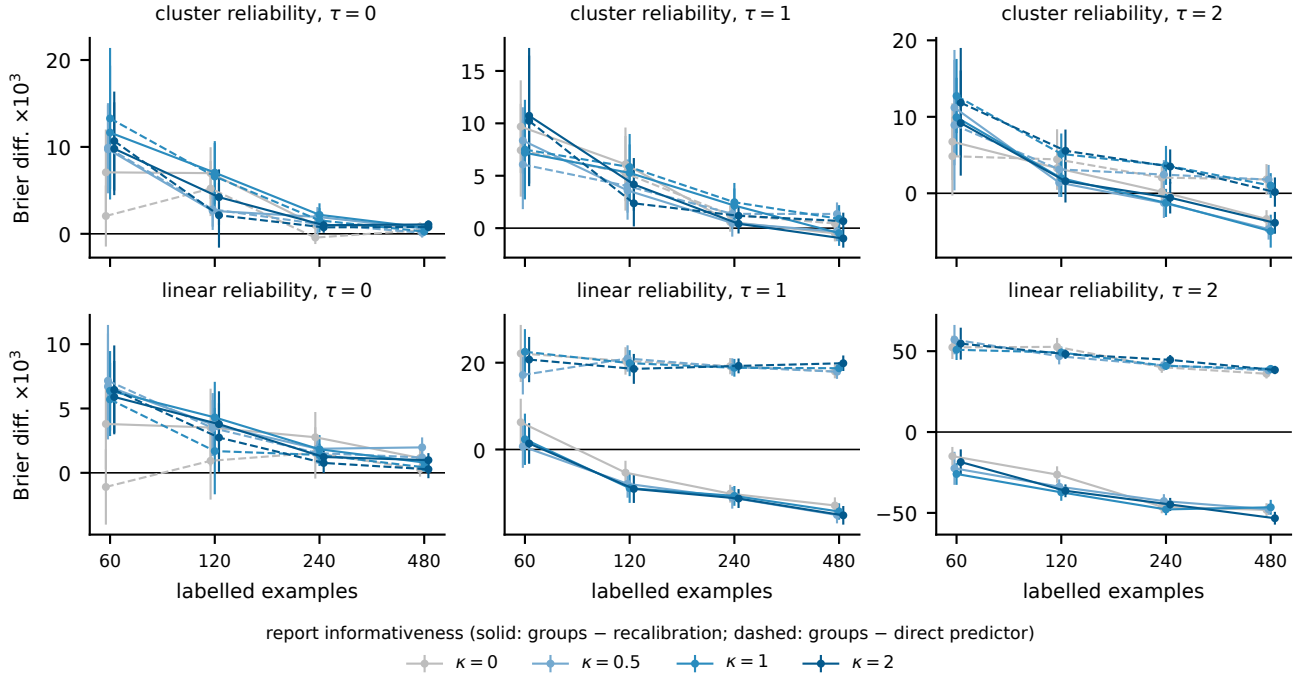


Figure 6: Synthetic reliability models. Brier difference ($\times 10^3$) of groups minus recalibration (solid) and groups minus the direct predictor (dashed), against labelled examples, for report informativeness κ (colour) and between-group spread τ (columns); 95% intervals over 30 independent seeds per configuration. Top: reliability varies across 32 clusters in four dimensions; bottom: linearly.

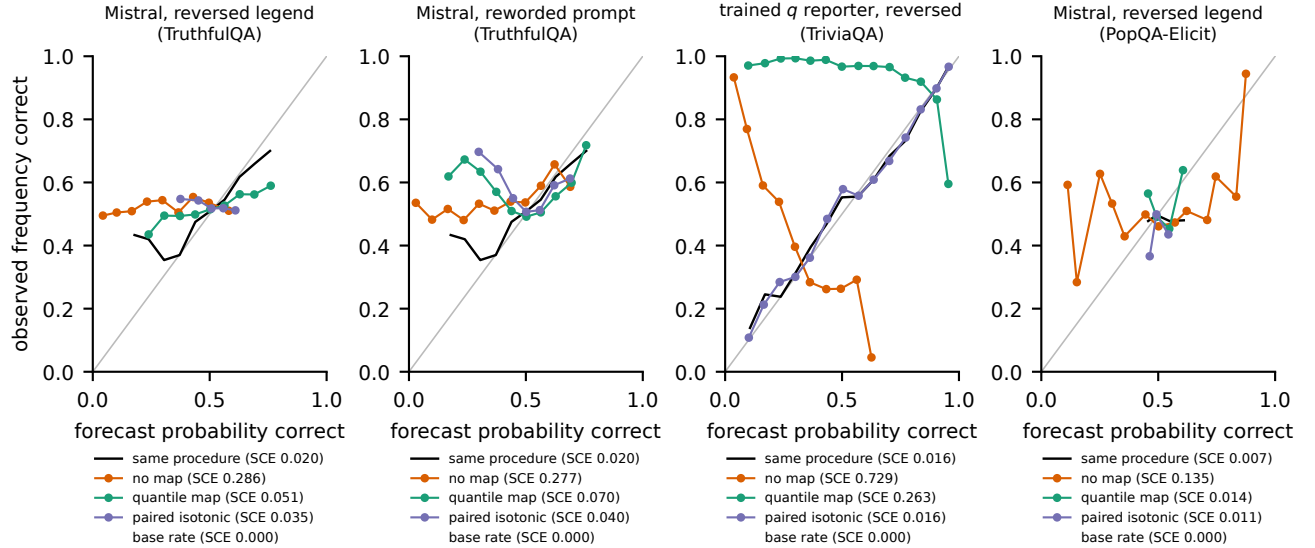


Figure 7: Reliability of a forecast fitted under procedure A and applied to procedure-B reports, with no map, the quantile map and paired isotonic regression, against the same forecast under A (black). Fifteen equal-width bins, bins with fewer than 20 predictions omitted; SCE from calibration-toolbox, with a constant forecast at the test base rate as reference (SCE 0 by construction). Test predictions pooled over 50 splits (TruthfulQA, POPQA-ELICIT) or the fixed split (TriviaQA). On PopQA Mistral’s forward report is uninformative, so every forecast stays near 0.5.

Table 13: Full LABEL-FREE-TRANSFER results on TruthfulQA (500 splits), POPQA-ELICIT (100 splits) and TriviaQA (one split): coverage at $\alpha = 0.1$, its spread, mean set size, Brier score of the mapped forecaster, crossing mass (two-sided bound on the coverage change), label-loss mass (one-sided bound; Proposition 4(a)), and the worst of the four frozen groups’ coverage gaps net of the same statistic for $A \rightarrow A$ (which is below zero without any shift). Forecaster and thresholds use disjoint halves of the labelled fold.

A $\rightarrow$ B	map	coverage	10-90% (frozen) / 95% CI (TriviaQA)	set size	Brier	crossing mass	label-loss mass	worst group, net of $A \rightarrow A$
TruthfulQA Llama fwd $\rightarrow$ reversed	same	0.919	[0.88, 0.95]	1.84	0.2544	0.000	0.000	+0.000
TruthfulQA Llama fwd $\rightarrow$ reversed	naive	0.930	[0.87, 0.98]	1.85	0.2514	0.208	0.102	+0.010
TruthfulQA Llama fwd $\rightarrow$ reversed	quantile	0.929	[0.88, 0.96]	1.85	0.2513	0.207	0.102	+0.010
TruthfulQA Llama fwd $\rightarrow$ reversed	paired iso.	0.993	[0.98, 1.00]	1.99	0.2520	0.157	0.006	+0.095
TruthfulQA Llama fwd $\rightarrow$ reversed	paired iso., groups	0.991	[0.97, 1.00]	1.98	0.2520	0.157	0.009	+0.089
TruthfulQA Llama fwd $\rightarrow$ reversed	labels 15+15	0.938	[0.85, 1.00]	1.86	0.2795	0.165	0.133	-0.061
TruthfulQA Llama fwd $\rightarrow$ reversed	labels 60+60	0.902	[0.85, 0.95]	1.76	0.2521	0.279	0.208	-0.151
TruthfulQA Llama fwd $\rightarrow$ reworded	same	0.919	[0.88, 0.95]	1.84	0.2544	0.000	0.000	+0.000
TruthfulQA Llama fwd $\rightarrow$ reworded	naive	0.858	[0.70, 1.00]	1.72	0.2557	0.316	0.223	-0.073
TruthfulQA Llama fwd $\rightarrow$ reworded	quantile	0.917	[0.87, 0.96]	1.83	0.2540	0.190	0.100	-0.004
TruthfulQA Llama fwd $\rightarrow$ reworded	paired iso.	0.972	[0.94, 1.00]	1.94	0.2527	0.156	0.027	+0.066
TruthfulQA Llama fwd $\rightarrow$ reworded	paired iso., groups	0.970	[0.94, 1.00]	1.94	0.2529	0.156	0.028	+0.063
TruthfulQA Llama fwd $\rightarrow$ reworded	labels 15+15	0.934	[0.86, 0.99]	1.87	0.2872	0.156	0.122	-0.019
TruthfulQA Llama fwd $\rightarrow$ reworded	labels 60+60	0.900	[0.84, 0.95]	1.80	0.2594	0.279	0.162	-0.075
TruthfulQA Mistral fwd $\rightarrow$ reversed	same	0.922	[0.88, 0.96]	1.79	0.2457	0.000	0.000	+0.000
TruthfulQA Mistral fwd $\rightarrow$ reversed	naive	0.595	[0.52, 0.66]	1.19	0.3595	0.728	0.721	-0.404
TruthfulQA Mistral fwd $\rightarrow$ reversed	quantile	0.897	[0.85, 0.94]	1.79	0.2554	0.321	0.168	-0.052
TruthfulQA Mistral fwd $\rightarrow$ reversed	paired iso.	0.994	[0.99, 1.00]	1.99	0.2524	0.212	0.007	+0.104
TruthfulQA Mistral fwd $\rightarrow$ reversed	paired iso., groups	0.992	[0.98, 1.00]	1.98	0.2520	0.214	0.012	+0.095
TruthfulQA Mistral fwd $\rightarrow$ reversed	labels 15+15	0.938	[0.86, 0.99]	1.88	0.2901	0.203	0.114	-0.034
TruthfulQA Mistral fwd $\rightarrow$ reversed	labels 60+60	0.901	[0.85, 0.95]	1.79	0.2584	0.417	0.167	-0.076
TruthfulQA Mistral fwd $\rightarrow$ reworded	same	0.922	[0.88, 0.96]	1.79	0.2457	0.000	0.000	+0.000
TruthfulQA Mistral fwd $\rightarrow$ reworded	naive	0.631	[0.56, 0.70]	1.27	0.3651	0.679	0.642	-0.356
TruthfulQA Mistral fwd $\rightarrow$ reworded	quantile	0.889	[0.84, 0.93]	1.79	0.2609	0.259	0.132	-0.049
TruthfulQA Mistral fwd $\rightarrow$ reworded	paired iso.	0.981	[0.96, 1.00]	1.96	0.2535	0.210	0.019	+0.079
TruthfulQA Mistral fwd $\rightarrow$ reworded	paired iso., groups	0.976	[0.95, 1.00]	1.95	0.2543	0.210	0.023	+0.068
TruthfulQA Mistral fwd $\rightarrow$ reworded	labels 15+15	0.937	[0.85, 0.99]	1.87	0.2887	0.200	0.114	-0.011
TruthfulQA Mistral fwd $\rightarrow$ reworded	labels 60+60	0.902	[0.85, 0.95]	1.81	0.2597	0.411	0.144	-0.076
TruthfulQA Qwen fwd $\rightarrow$ reversed	same	0.919	[0.88, 0.96]	1.80	0.2527	0.000	0.000	+0.000
TruthfulQA Qwen fwd $\rightarrow$ reversed	naive	0.826	[0.61, 1.00]	1.65	0.2587	0.391	0.280	-0.120
TruthfulQA Qwen fwd $\rightarrow$ reversed	quantile	0.892	[0.83, 0.95]	1.79	0.2574	0.230	0.123	-0.040
TruthfulQA Qwen fwd $\rightarrow$ reversed	paired iso.	0.940	[0.80, 1.00]	1.88	0.2549	0.213	0.071	+0.021
TruthfulQA Qwen fwd $\rightarrow$ reversed	paired iso., groups	0.954	[0.84, 1.00]	1.91	0.2562	0.202	0.050	+0.038
TruthfulQA Qwen fwd $\rightarrow$ reversed	labels 15+15	0.940	[0.86, 0.99]	1.88	0.2857	0.147	0.116	-0.037
TruthfulQA Qwen fwd $\rightarrow$ reversed	labels 60+60	0.902	[0.85, 0.95]	1.80	0.2577	0.304	0.175	-0.088
TruthfulQA Qwen fwd $\rightarrow$ reworded	same	0.919	[0.88, 0.96]	1.80	0.2527	0.000	0.000	+0.000
TruthfulQA Qwen fwd $\rightarrow$ reworded	naive	0.949	[0.83, 1.00]	1.88	0.2506	0.222	0.072	+0.035
TruthfulQA Qwen fwd $\rightarrow$ reworded	quantile	0.913	[0.87, 0.96]	1.78	0.2487	0.205	0.114	-0.009
TruthfulQA Qwen fwd $\rightarrow$ reworded	paired iso.	0.948	[0.81, 1.00]	1.88	0.2508	0.203	0.066	+0.029
TruthfulQA Qwen fwd $\rightarrow$ reworded	paired iso., groups	0.964	[0.92, 1.00]	1.91	0.2527	0.183	0.039	+0.054
TruthfulQA Qwen fwd $\rightarrow$ reworded	labels 15+15	0.935	[0.86, 0.99]	1.84	0.2790	0.178	0.147	-0.031
TruthfulQA Qwen fwd $\rightarrow$ reworded	labels 60+60	0.899	[0.84, 0.95]	1.74	0.2496	0.303	0.219	-0.120
PopQA Llama fwd $\rightarrow$ reversed	same	0.901	[0.89, 0.91]	1.73	0.2445	0.000	0.000	+0.000
PopQA Llama fwd $\rightarrow$ reversed	naive	0.596	[0.56, 0.64]	1.12	0.2564	0.812	0.768	-0.324
PopQA Llama fwd $\rightarrow$ reversed	quantile	0.951	[0.94, 0.96]	1.72	0.2255	0.385	0.211	+0.040
PopQA Llama fwd $\rightarrow$ reversed	paired iso.	1.000	[1.00, 1.00]	2.00	0.2454	0.268	0.003	+0.124
PopQA Llama fwd $\rightarrow$ reversed	paired iso., groups	0.998	[1.00, 1.00]	1.98	0.2454	0.264	0.009	+0.116
PopQA Llama fwd $\rightarrow$ reversed	labels 15+15	0.946	[0.88, 0.99]	1.75	0.2373	0.256	0.235	+0.039
PopQA Llama fwd $\rightarrow$ reversed	labels 60+60	0.899	[0.84, 0.95]	1.56	0.2138	0.444	0.412	-0.025
PopQA Llama fwd $\rightarrow$ reworded	same	0.901	[0.89, 0.91]	1.73	0.2445	0.000	0.000	+0.000
PopQA Llama fwd $\rightarrow$ reworded	naive	0.905	[0.87, 0.94]	1.74	0.2444	0.387	0.200	-0.041
PopQA Llama fwd $\rightarrow$ reworded	quantile	0.902	[0.88, 0.92]	1.72	0.2430	0.392	0.210	-0.041
PopQA Llama fwd $\rightarrow$ reworded	paired iso.	0.996	[0.99, 1.00]	1.99	0.2481	0.266	0.006	+0.114
PopQA Llama fwd $\rightarrow$ reworded	paired iso., groups	0.979	[0.97, 0.99]	1.94	0.2464	0.252	0.025	+0.081
PopQA Llama fwd $\rightarrow$ reworded	labels 15+15	0.932	[0.85, 0.99]	1.84	0.2818	0.182	0.156	-0.013
PopQA Llama fwd $\rightarrow$ reworded	labels 60+60	0.902	[0.85, 0.95]	1.75	0.2517	0.330	0.222	-0.057
PopQA Mistral fwd $\rightarrow$ reversed	same	0.901	[0.89, 0.91]	1.80	0.2505	0.000	0.000	+0.000
PopQA Mistral fwd $\rightarrow$ reversed	naive	0.561	[0.49, 0.91]	1.12	0.2721	0.799	0.775	-0.344
PopQA Mistral fwd $\rightarrow$ reversed	quantile	0.842	[0.78, 0.92]	1.70	0.2517	0.411	0.260	-0.176
PopQA Mistral fwd $\rightarrow$ reversed	paired iso.	0.965	[0.90, 1.00]	1.92	0.2499	0.235	0.056	+0.006
PopQA Mistral fwd $\rightarrow$ reversed	paired iso., groups	0.996	[0.99, 1.00]	1.99	0.2502	0.201	0.005	+0.107
PopQA Mistral fwd $\rightarrow$ reversed	labels 15+15	0.928	[0.84, 0.99]	1.73	0.2511	0.267	0.266	+0.024
PopQA Mistral fwd $\rightarrow$ reversed	labels 60+60	0.906	[0.85, 0.95]	1.64	0.2256	0.365	0.364	-0.001
PopQA Mistral fwd $\rightarrow$ reworded	same	0.901	[0.89, 0.91]	1.80	0.2505	0.000	0.000	+0.000
PopQA Mistral fwd $\rightarrow$ reworded	naive	0.572	[0.50, 0.87]	1.14	0.2713	0.785	0.751	-0.342
PopQA Mistral fwd $\rightarrow$ reworded	quantile	0.874	[0.82, 0.95]	1.77	0.2516	0.275	0.154	-0.090
PopQA Mistral fwd $\rightarrow$ reworded	paired iso.	0.943	[0.88, 1.00]	1.89	0.2509	0.230	0.071	-0.010
PopQA Mistral fwd $\rightarrow$ reworded	paired iso., groups	0.985	[0.97, 1.00]	1.97	0.2508	0.197	0.013	+0.091
PopQA Mistral fwd $\rightarrow$ reworded	labels 15+15	0.929	[0.84, 1.00]	1.77	0.2579	0.234	0.233	+0.022
PopQA Mistral fwd $\rightarrow$ reworded	labels 60+60	0.900	[0.84, 0.94]	1.65	0.2330	0.349	0.348	-0.016
PopQA Qwen fwd $\rightarrow$ reversed	same	0.901	[0.89, 0.91]	1.59	0.2193	0.000	0.000	+0.000
PopQA Qwen fwd $\rightarrow$ reversed	naive	0.526	[0.50, 0.56]	1.02	0.3026	0.760	0.748	-0.389
PopQA Qwen fwd $\rightarrow$ reversed	quantile	0.896	[0.88, 0.91]	1.58	0.2202	0.360	0.195	-0.036
PopQA Qwen fwd $\rightarrow$ reversed	paired iso.	0.876	[0.83, 0.92]	1.61	0.2351	0.390	0.196	-0.190
PopQA Qwen fwd $\rightarrow$ reversed	paired iso., groups	0.882	[0.83, 0.93]	1.61	0.2281	0.350	0.170	-0.144
PopQA Qwen fwd $\rightarrow$ reversed	labels 15+15	0.928	[0.83, 0.99]	1.73	0.2463	0.285	0.181	+0.002
PopQA Qwen fwd $\rightarrow$ reversed	labels 60+60	0.899	[0.86, 0.95]	1.61	0.2246	0.371	0.206	-0.034
PopQA Qwen fwd $\rightarrow$ reworded	same	0.901	[0.89, 0.91]	1.59	0.2193	0.000	0.000	+0.000
PopQA Qwen fwd $\rightarrow$ reworded	naive	0.815	[0.73, 0.88]	1.53	0.2532	0.433	0.263	-0.330
PopQA Qwen fwd $\rightarrow$ reworded	quantile	0.894	[0.88, 0.91]	1.58	0.2236	0.302	0.162	-0.092
PopQA Qwen fwd $\rightarrow$ reworded	paired iso.	0.907	[0.88, 0.94]	1.70	0.2292	0.316	0.109	-0.188
PopQA Qwen fwd $\rightarrow$ reworded	paired iso., groups	0.914	[0.88, 0.95]	1.69	0.2271	0.280	0.094	-0.102
PopQA Qwen fwd $\rightarrow$ reworded	labels 15+15	0.935	[0.85, 0.99]	1.74	0.2487	0.287	0.151	+0.023
PopQA Qwen fwd $\rightarrow$ reworded	labels 60+60	0.898	[0.85, 0.95]	1.60	0.2238	0.342	0.149	-0.017
TriviaQA question-only	same	0.909	[0.90, 0.91]	1.20	0.1228	0.000	0.000	-
TriviaQA question-only	naive	0.206	[0.20, 0.21]	1.05	0.7099	0.973	0.953	-
TriviaQA question-only	quantile	0.780	[0.77, 0.79]	1.18	0.2392	0.442	0.258	-
TriviaQA question-only	paired iso.	0.910	[0.90, 0.92]	1.20	0.1232	0.024	0.013	-
TriviaQA question-only	labels 15+15	0.969	-	1.50	0.1406	0.502	0.502	-
TriviaQA question-only	labels 60+60	0.893	-	1.16	0.1280	0.065	0.063	-
TriviaQA answer-aware	same	0.910	[0.90, 0.92]	1.13	0.1043	0.000	0.000	-
TriviaQA answer-aware	naive	0.460	[0.45, 0.47]	1.27	0.4460	0.919	0.651	-
TriviaQA answer-aware	quantile	0.775	[0.77, 0.78]	1.12	0.1884	0.354	0.277	-
TriviaQA answer-aware	paired iso.	0.982	[0.98, 0.98]	1.51	0.1440	0.381	0.003	-
TriviaQA answer-aware	labels 15+15	0.984	-	1.58	0.1599	0.252	0.007	-
TriviaQA answer-aware	labels 60+60	0.885	-	1.27	0.1450	0.336	0.128	-

Table 14: Pre-set E5 analysis (superseded): Brier difference against R with naive intervals, fixed penalty. Bold: worse than R.

Brier difference vs. R (95% CI)	$n_{\text{lab}} = 60$	120	180	272
Llama, G_4	-.003	-.007	-.008	-.009
Mistral, G_4	+. 004	-.002	-.004	-.005
Qwen, G_4	-.004	-.008	-.009	-.010
G_8 (three models)	+.005 to +.013			-.007 to -.011
D (three models)	+.005 to +.010	-.014 to -.019		-.030 to -.034
MC (three models)	+.000 to +.002			-.004 to +.003
Brier-minimising M (Llama / Mistral / Qwen)	4 / 1 / 4	4 / 4 / 4	4 / 8 / 4	16 / 8 / 8